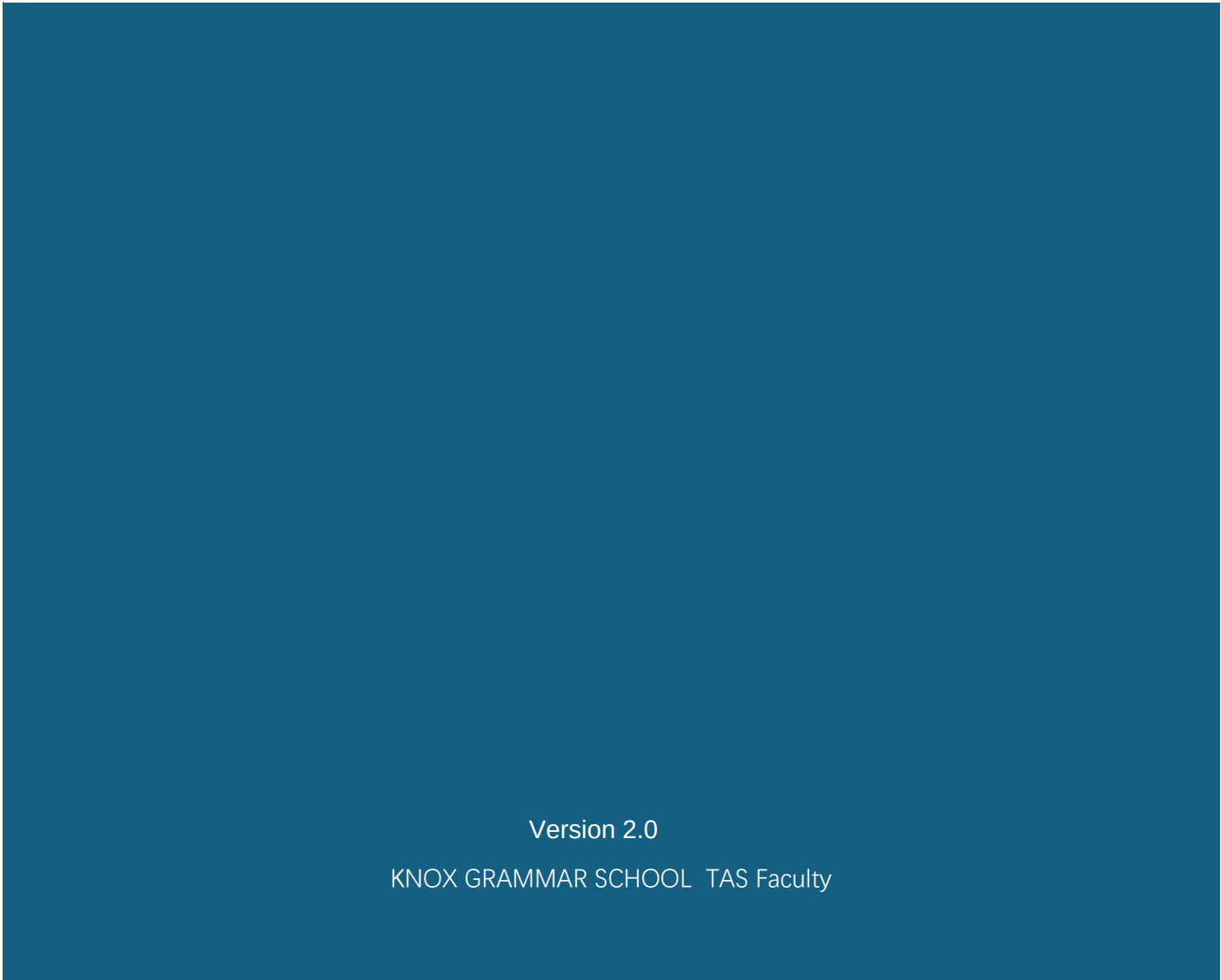




SOFTWARE ENGINEERING

Version 2.0

KNOX GRAMMAR SCHOOL TAS Faculty



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How to Use This Book

This resource has been compiled to support Year 12 Software Engineering students at Knox Grammar School in preparing for the HSC examination. It brings together the **2025 HSC Software Engineering Familiarisation Questions**, the **2025 HSC Software Engineering Examination**, and a **Sample Examination**, each accompanied by their official NESA Marking Guidelines. Working through these materials will build familiarity with examination question styles, marking criteria, and the depth of response expected at HSC level.

Examination Overview

The following table summarises the key specifications for the HSC Software Engineering examination as published by NESA.

General Examination Details	
Total marks	80 marks
Time allowed	2 hours and 30 minutes (including 10 minutes reading time)
Format	Computer-based examination
Assumed knowledge	Year 11 Software Engineering course content and Python programming language
Stimulus material	Questions may contain stimulus material including text, images, code, and data
Headphones required	Required for questions with video and/or audio stimulus
Question Types	
Objective-response items	Approximately 20 marks total 14–18 items in total Each item worth 1 to 4 marks Items may share stimulus with other items
Short-answer items	Approximately 60 marks total 16–18 items in total At least three items worth 4 to 8 marks Items may share stimulus with other items

Source: NESA HSC Software Engineering Examination Specifications, Stage 6 — Year 12.
<https://curriculum.nsw.edu.au/learning-areas/tas/software-engineering-11-12-2022/assessment>

Important Notes for Students

Keep the following points in mind as you work through this resource:

- Examination questions may require you to integrate knowledge, understanding, and skills developed across the entire course — not just a single topic.
- The Year 11 course is assumed knowledge for the Year 12 course. Ensure your understanding of Year 11 content is solid before attempting harder questions.
- A working understanding of Python is assumed. Practice reading, tracing, and writing Python code regularly.

- Reading time (10 minutes) should be used strategically — read the entire paper, identify high-mark questions, and plan your approach before writing.
- Headphones will be required during the examination for any questions that include video or audio stimulus material.

How This Book Is Organised

This book is divided into three sections, each containing an examination paper followed immediately by its NESA Marking Guidelines.

Section	Content	Purpose
1	2025 HSC Familiarisation Questions & Marking Guidelines	Released by NESA to introduce students to the new computer-based examination format. Covers question styles including ordering, matching, checkbox, dropdown, and extended response. Ideal starting point.
2	2025 HSC Sample Examination & Marking Guidelines	A full-length practice examination set by NESA to model what the actual HSC examination will look like. Use this for timed practice under exam conditions.
3	2025 HSC Examination & Marking Guidelines	The actual 2025 HSC examination. Study the marking guidelines carefully to understand how marks are awarded and what examiners look for.

How to Use the Marking Guidelines

The Marking Guidelines are one of the most valuable study tools available. Here is how to get the most from them:

- Attempt the questions first before looking at the answers. Resist the temptation to read the guidelines immediately.
- For objective-response questions, check whether your answer is correct and identify any misunderstandings.
- For short-answer questions, compare your response against the criteria table. Ask yourself: did I address every marking criterion? Did I use the correct command verb (identify, outline, describe, explain, discuss, evaluate)?
- Study sample answers carefully. Note the structure, level of detail, and use of terminology — not just the content.
- Use the Mapping Grid (at the end of each marking guidelines document) to identify which syllabus outcomes and content areas you are weakest in, then target those for revision.

Understanding Command Verbs

Short-answer questions use specific command verbs that tell you exactly how much depth is required. Understanding these will help you write responses that match the marking criteria.

Verb	What is expected
Identify	Name or select a feature, fact, or example. No explanation required.
Outline	Give a brief account of the main features. More than just naming — a sentence or two per point.
Describe	Provide characteristics and features in detail. Show what something is like.
Explain	Make clear how or why something works. Show cause and effect or reasoning.
Discuss	Provide points for and/or against a topic. Consider multiple perspectives.
Evaluate / Assess	Make a judgement based on evidence. Justify your conclusion.
Compare	Show how things are similar and/or different. Both sides must be addressed.

Suggested Study Approach

To get the most out of this resource, consider the following approach:

- **Begin with the Familiarisation Questions** (Section 1) to become comfortable with the computer-based question formats before attempting full papers.
- **Attempt the Sample Examination** (Section 2) under timed conditions (2 hours 20 minutes of writing time). Use the Mapping Grid to identify gaps.
- **Attempt the 2025 HSC Examination** (Section 3) as your final full mock. Compare your responses against the official marking guidelines and sample answers.
- **Revisit any outcome areas** where you consistently score below full marks. Use your course notes and the syllabus to fill those gaps.

This resource has been compiled for educational purposes using publicly available NESA examination and marking guidelines materials.

HSC Software Engineering

Familiarisation Questions

Question FAM 1 (1 mark)

Place the following software development steps in the correct order by clicking and holding each step and dragging it into its position.

Steps to Order	Your Order (1 to 6)
Design	
Installation	
Maintenance	
Development	
Testing and debugging	
Determining specifications	

Question FAM 2 (3 marks)

An online business is planning to use a database to keep track of its products. The contents of the Products table during testing are shown.

Products		
ProductID	ProductName	ProductPrice
P001	The Plant (DVD)	28.00
P002	Discovery 1 and 2 (DVD)	26.98
P003	Travel 1 and 2 (Blu-Ray)	22.00
P004	The Best Movie (DVD)	19.98
P005	Celebration (DVD)	12.00

After executing a SQL query, the following results were obtained.

ProductName	ProductPrice
Discovery 1 and 2 (DVD)	26.98
The Plant (DVD)	28.00
Travel 1 and 2 (Blu-Ray)	22.00

Select the correct item from each dropdown menu to show how the results were obtained.

SQL Query structure (Circle the correct dropdown item):

```
SELECT [dropdown: Products ProductID ProductName ProductPrice]
[dropdown: Products ProductID ProductName ProductPrice]
```

```

FROM      [dropdown: Products ProductID ProductName ProductPrice]
WHERE     [dropdown: Products ProductID ProductName
ProductPrice][dropdown: < = >= <= <> ] [dropdown: 12.00 19.98 22.00
26.98 28.00]
ORDER BY  [dropdown: Products ProductID ProductName ProductPrice]
[dropdown: ASC DESC]

```

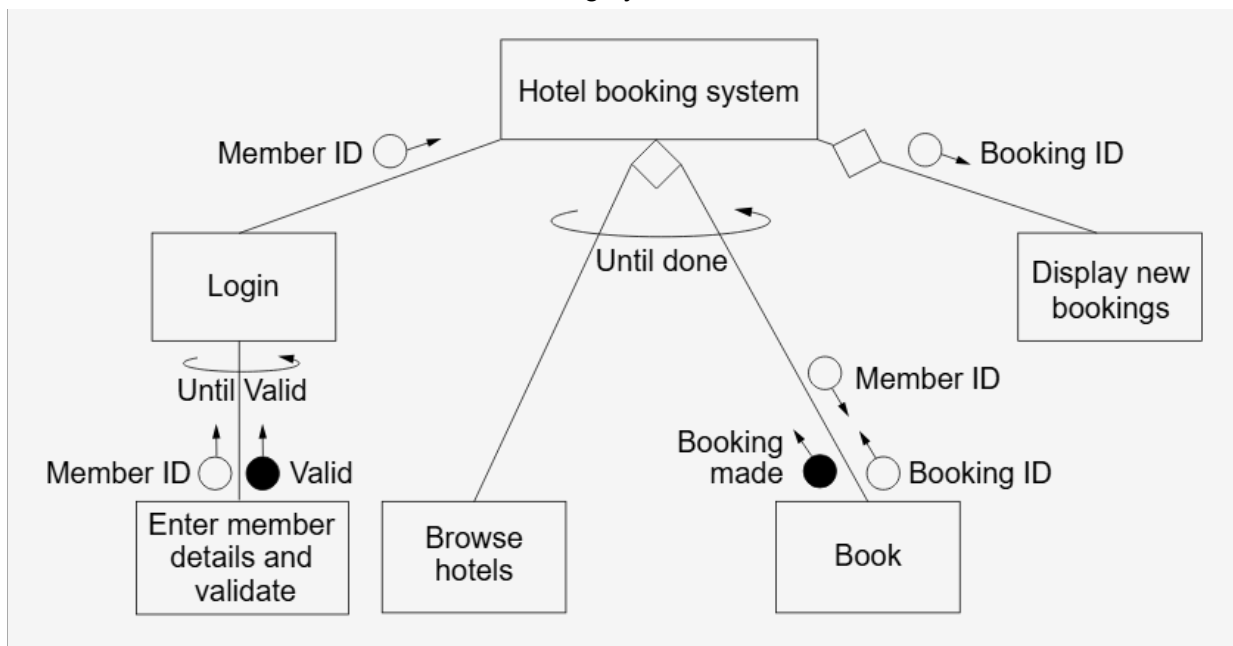
Question FAM 3 (2 marks)

Match each of the features to the relevant protocol(s).

	POP3	SMTP	IMAP
Commonly used for sending email	☐	☐	☐
Commonly used for receiving email	☐	☐	☐
Allows email to synchronise across devices	☐	☐	☐
Server deletes messages when they are downloaded and opened	☐	☐	☐

Question FAM 4 (2 marks)

This structure chart describes a hotel booking system.



Select the items that are TRUE about this structure chart. (Multiple items may be selected.)

- ☐ **Booking made** is a flag.
- ☐ **Login** is the first module called.
- ☐ **Booking ID** may only be true or false.
- ☐ **Display new bookings** may not be executed.
- ☐ Members do not need to log in to browse hotels.
- ☐ A choice is continually offered between **Browse hotels** and **Book**.
- ☐ **Login** calls **Enter member details and validate** when Member ID is set to true.

Question FAM 5 (3 marks)

Consider the following algorithm.

```

BEGIN determineGrade
  get Mark
  CASEWHERE Mark
    >=80 : Grade = "H"
    >=50 : Grade = "P"
    OTHERWISE : Grade = "N"
  ENDCASE
  print Grade
End determineGrade

```

Use the following table to provide a set of test data that will thoroughly test the algorithm. Include the expected outputs and reasons for inclusion. You may assume that all input data are valid.

Test data (Mark)	Expected output (Grade)	Reason for inclusion

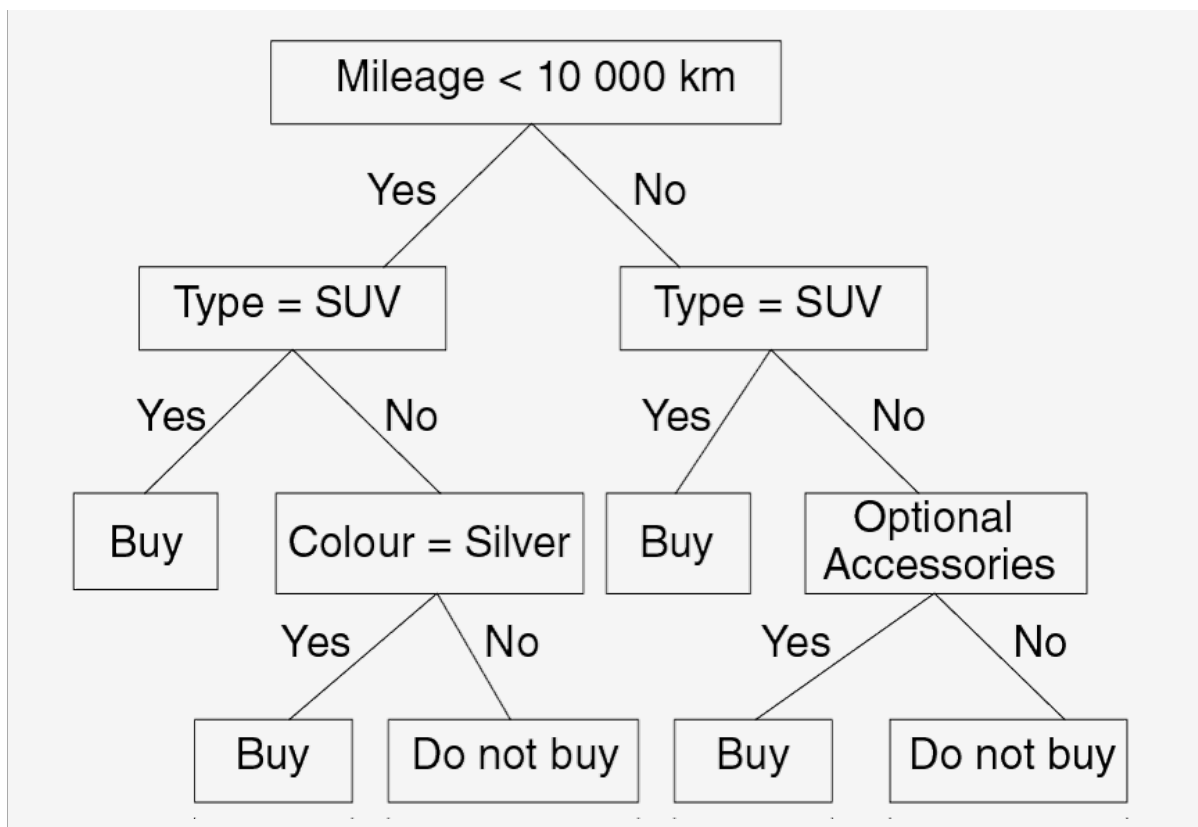
Question FAM 6 (3 marks)

A new operating system is to be developed for a mobile phone. The operating system needs to allow users to install a variety of third-party applications such as social media applications and games on the phone.

Discuss the use of sandboxing in this operating system.

Question FAM 7 (4 marks)

Consider the following decision tree of a trained machine learning model that determines whether to purchase a vehicle.



(a) Using the decision tree, determine the outcome of each of the following situations. (1 mark)

	Buy	Do not buy
Mileage = 8000 km, Colour = Silver, Optional Accessories = No	☐	☐
Mileage = 11 000 km, Colour = Red, Optional Accessories = Yes	☐	☐
Type = SUV, Colour = Red, Optional Accessories = No	☐	☐

(b) The decision tree can be simplified without compromising its logic. Redraw the decision tree to reduce the number of branches. (3 marks)

Question FAM 8 (6 marks)

The following subroutine is intended to search an array for an item and output the position of that item in the array. There are several mistakes in the algorithm.

```
BEGIN Search (SearchItem)
    Set LastIndex to the number of elements in ItemArray
    Index = 1
    WHILE Found = FALSE OR Index < LastIndex
        IF Index = SearchItem THEN
            Found = TRUE
        ENDIF
    ENDWHILE
    IF Found = TRUE THEN
        Display 'Position', Index
    ELSE
        Display 'Not found'
    ENDIF
END Search
```

(a) Copy the algorithm into the space below and modify the algorithm logic so that it achieves its intended purpose. (The array is called ItemArray and it is indexed from 1.) **(3 marks)**

Question FAM 9 (6 marks)

An online store allows users to register for an account using their name, phone number and date of birth. Users must also create a username and password.

Each username must satisfy these rules:

- There must be no more than 8 characters.
- Only uppercase and lowercase letters are accepted.
- The character "<" is not allowed, to prevent attempts at injecting code through the username.

(a) Design a user interface for registering an account. Clearly label all features. **(3 marks)**



(b) Write a function in Python that will check whether a username satisfies the rules. **(3 marks)**

Question FAM 10 (4 marks)

Consider the following source code for a web page.

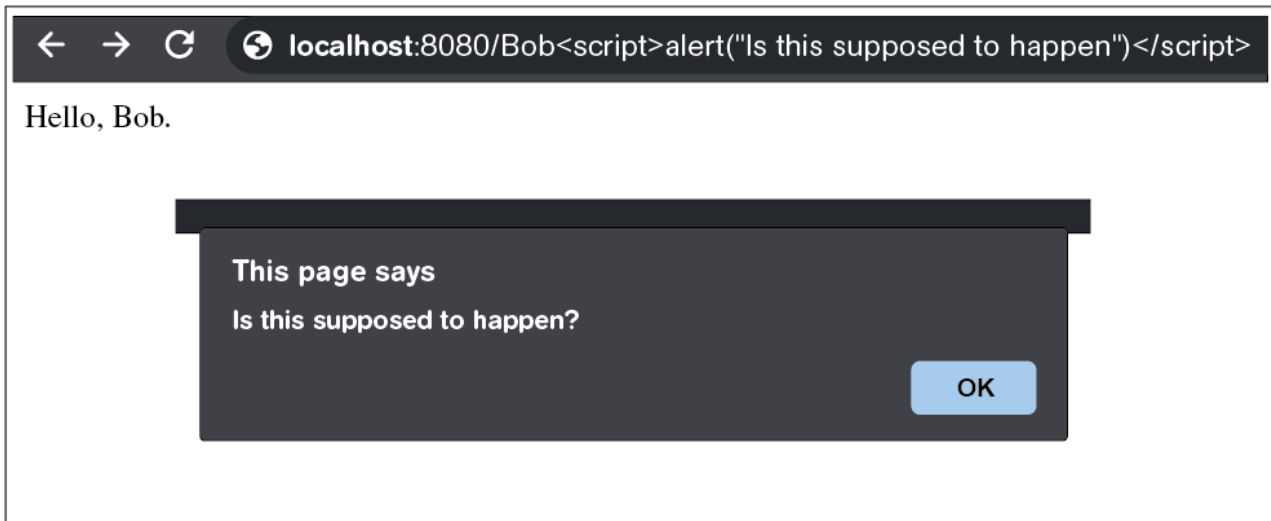
```
<html>
<body>
  <link rel="stylesheet" href="style3.css"/>
  <script src="image-gallery.js"/><!--this script is slow to load-->
  <link rel="stylesheet" href="styles1.css"/>
  <link rel="stylesheet" href="styles2.css"/>
  <h1>My Awesome Website!</h1>
  <p>I hope you like my website, I spent a lot of time designing
it.</p>
  <h2>Photos from my holiday</h2>
  
  
  
  
</body>
</html>
```

The web page is loading too slowly. Explain how the load time of this page can be improved.

Question FAM 11 (3 marks)

Bob is testing a website for vulnerability. The website allows the user to add their name at the end of the URL and prints their name on the webpage.

In addition to his name, Bob has added some code to the URL and malicious output is displayed on the page.



(a) What type of vulnerability is being demonstrated? (1 mark)

- ☐ Invalid redirecting
- ☐ Cross site scripting
- ☐ Broken authentication
- ☐ Cross site request forgery

(b) Explain ONE way to minimise this vulnerability. (2 marks)

Question FAM 12 (5 marks)

An online business is planning to use a database to keep track of its products, customers and orders. The contents of the Products, Customers and Orders tables during testing are shown.

Products

ProductID	ProductName	ProductPrice
P001	The Plant (DVD)	28.00
P002	Discovery 1 and 2 (DVD)	26.98
P003	Travel 1 and 2 (Blu-Ray)	22.00
P004	The Best Movie (DVD)	19.98
P005	Celebration (DVD)	12.00

Customers

CustID	CustName	CustEmail
C001	Nicky Singh	N.Singh@nesa.edu.au
C002	Casey He	C.He@bos.edu.au
C003	Ash Lee	A.Lee@bosnsw.edu
C004	Kim Smith	K.Smith@nesa.edu.au
C005	Pat Alvarado	P.Alvarado@bos.edu.au

Orders

OrderID	CustID	ProductID	OrderQuantity
T001	C001	P004	2
T002	C004	P005	1
T003	C002	P003	1
T004	C001	P002	2
T005	C005	P002	2

(a) The following SQL query is run to test the database. (2 marks)

```
SELECT ProductName, ProductPrice
FROM Products
WHERE ProductPrice > 22.00
ORDER BY ProductPrice ASC
```

Fill in the table below with the result of running the SQL query. You may not need to use all the rows or columns.

(b) The contents of the Orders table need to be displayed showing customer names and their ordered products. Write the SQL query to achieve this. **(3 marks)**

Customer	Product	Quantity
Nicky Singh	The Best Movie (DVD)	2
Kim Smith	Celebration (DVD)	1
Casey He	Travel 1 and 2 (Blu-Ray)	1
Nicky Singh	Discovery 1 and 2 (DVD)	2
Pat Alvarado	Discovery 1 and 2 (DVD)	2

Question FAM 13 (6 marks)

In a computer game, each player has 10 rounds and the maximum score for each round is 5. The score for each round is equal to the number of points, except when bonus points are assigned. Every time a player scores 5, the score from the next round is added as bonus points. If a player scores 5 in the last round, they get 10 points.

A player's scores and corresponding points for a full game are shown.

Round	1	2	3	4	5	6	7	8	9	10
Score	3	5	5	2	1	4	0	1	4	5
Points	3	10	7	2	1	4	0	1	4	10

Write a program in Python that will:

- display the player's scores and corresponding points

- calculate and display the total points for the player, taking into account the bonuses.

Start your program with: `scores = [3,5,5,2,1,4,0,1,4,5]`

The output of the program should match the expected format.

Round	Score	Points
1	3	3
2	5	10
3	5	7
4	2	2
5	1	1
6	4	4
7	0	0
8	1	1
9	4	4
10	5	10
TOTAL POINTS:		42

Question FAM 14 (8 marks)

The following slideshow was prepared by a student as part of a research project. (12 slides on artificial intelligence in society)

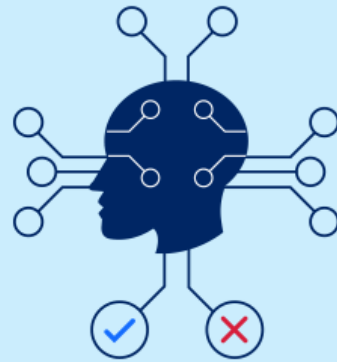


ARTIFICIAL INTELLIGENCE

How many times have you
interacted with artificial
intelligence this week?



How many decisions have been made about **YOU** this week by artificial intelligence?

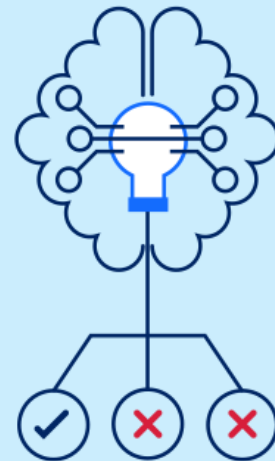


AI can notice patterns in enormous data sets that are beyond the ability of human perception to detect.

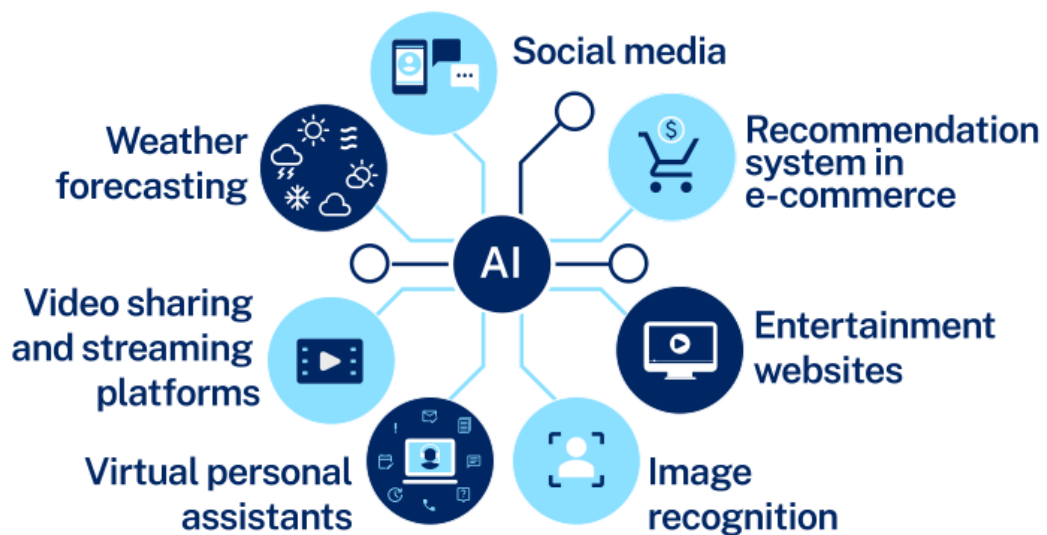
AI technology can make even ordinary and seemingly trivial data valuable.



AI can facilitate decision-making and improve productivity.



Artificial intelligence is used in areas such as:





AI robots
can take on
risky or
boring tasks.



**AI medical
assistants**
can diagnose
diseases based
on patients'
symptoms.

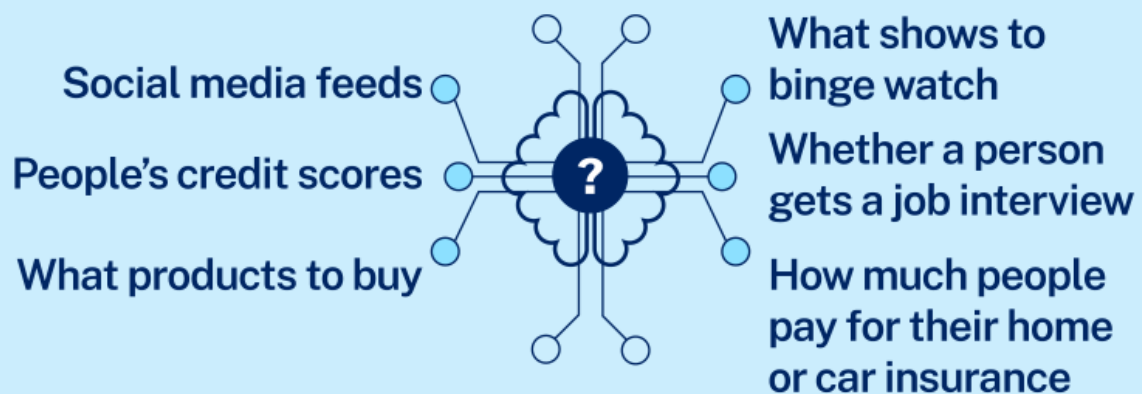


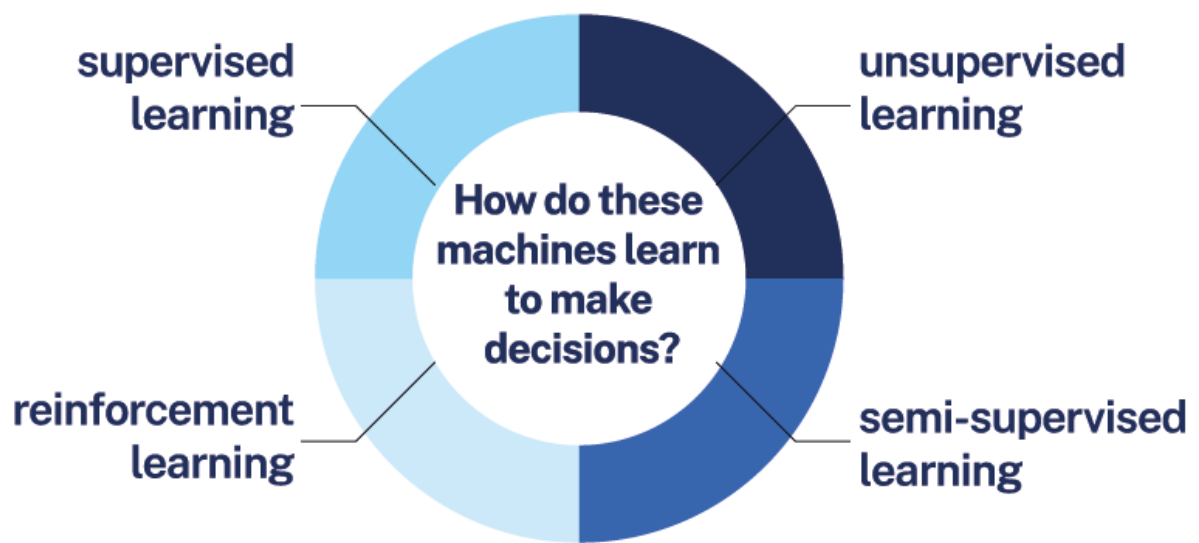
Chatbots
can answer
customer service
questions.



AI
can even filter
out hostile
activities
to protect
cyber security.

**Algorithms are being used all
the time to make decisions about . . .**





AI needs **LARGE** amounts of training data.

Big data is the raw material that fuels AI algorithms and models.



How do they learn about us?

Social media

Online surveys

Shopping
history

Internet browsing
history and
preference



Our everyday
interactions
with technology



**What will our future
relationship with AI be?**

Assess the use of artificial intelligence in our society. Support your answer with information from the slideshow.

— End of Familiarisation Examination —

2025 HSC Software Engineering Familiarisation Questions Marking Guidelines

Question 1

Criteria	Marks
• Correctly orders the steps	1

Answer:

≡ Determining specifications

≡ Design

≡ Development

≡ Testing and debugging

≡ Installation

≡ Maintenance

Question 2

Criteria	Marks
• Correctly selects all options	3
• Correctly selects at least FIVE options	2
• Correctly selects at least TWO options	1

Answer:

SELECT	ProductName ▾	ProductPrice ▾
FROM	Products ▾	
WHERE	ProductPrice ▾	>= ▾ 22.00 ▾
ORDER BY	ProductName ▾	ASC ▾

Question 3

Criteria	Marks
• Correctly matches the features to the relevant protocols	2
• Correctly matches at least TWO features to the relevant protocols	1

Answer:

	POP3	SMTP	IMAP
Commonly used for sending email	☐	☒	☐
Commonly used for receiving email	☒	☐	☒
Allows email to synchronise across devices	☐	☐	☒
Server deletes messages when they are downloaded and opened	☒	☐	☐

Question 4

Criteria	Marks
• Correctly identifies all the items that are true about the structure chart	2
• Identifies true/false correctly for at least FIVE checkboxes	1

Sample answer:

- ☒ **Booking made** is a flag.
- ☒ **Login** is the first module called.
- ☐ **Booking ID** may only be true or false.
- ☒ **Display new bookings** may not be executed.
- ☐ Members do not need to log in to browse hotels.
- ☒ A choice is continually offered between **Browse hotels** and **Book**.
- ☐ **Login** calls **Enter member details and validate** when Member ID is set to true.

Question 5

Criteria	Marks
- Provides a set of test data within the table that will thoroughly test the algorithm - Includes expected outputs and reasons for inclusion	3
- Provides a set of test data that tests most boundaries and/or paths - Includes most of the expected outputs and reasons for inclusion	2
Provides some relevant information	1

Sample answer:

Test data (Mark)	Expected output (Grade)	Reason for inclusion
85	H	Testing greater than 80
80	H	Testing boundary value (80)
60	P	Testing between 50 and 80
50	P	Testing boundary value (50)
45	N	Testing 'otherwise' (below 50)

Question 6

Criteria	Marks
• Provides a thorough discussion of the use of <i>sandboxing</i> in the operating system	3
• Shows some understanding of the advantages and/or disadvantages of the use of <i>sandboxing</i>	2
• Identifies a feature of <i>sandboxing</i>	1

Sample answer:

Sandboxing allows each application to be isolated so that it cannot pose a security risk by accessing system resources or other applications. However, it requires more resources as each application requires its own container. It may also be less convenient for users to access resources such as the device's camera or files as the sandbox will either prevent access or make it more difficult.

Answers could include:

Sandboxing limits the resource use of different processes (eg applications) running on the device, ensuring that if a process contains bad code, such as a memory leak, it does not consume all the resources of the operating system – potentially starving other processes, or the operating system itself, of resources.

Question 7 (a)

Criteria	Marks
• Correctly determines the outcomes	1

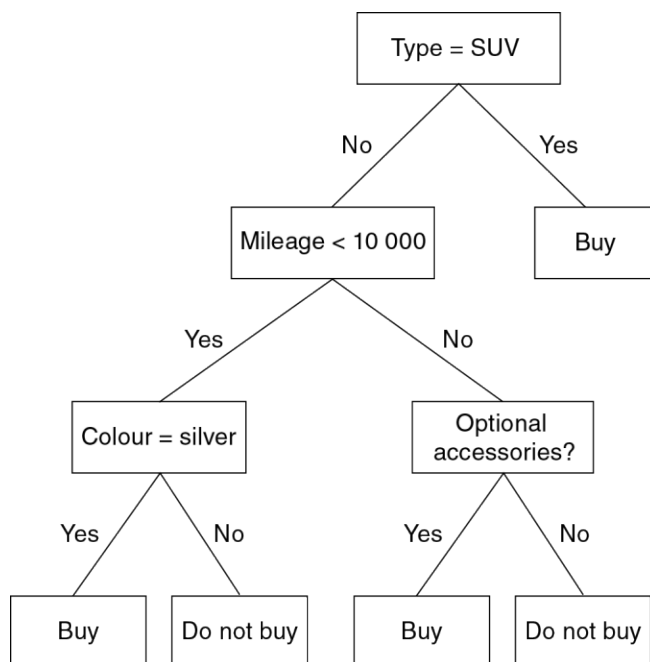
Answer:

	Buy	Not Buy
Mileage = 8000 km, Colour = Silver, Optional Accessories = No	☒	☐
Mileage = 10 000 km, Colour = Red, Optional Accessories = Yes	☒	☐
Type = SUV, Colour = Red, Optional Accessories = No	☒	☐

Question 7 (b)

Criteria	Marks
• Draws a correctly simplified decision tree	3
• Draws a simplified decision tree with most of the logic	2
• Shows some understanding of the requirement	1

Sample answer:



Question 8

Criteria	Marks
• Correctly modifies the algorithm to achieve its intended purpose	3
• Corrects some mistakes in the algorithm	2
• Shows some understanding of the problem	1

Sample answer:

```
BEGIN Search (SearchItem)
  Set LastIndex to the number of elements in ItemArray
  Found = FALSE
  Index = 1
  WHILE Found = FALSE AND Index < LastIndex + 1
    IF ItemArray(Index) = SearchItem THEN
      Found = TRUE
    ENDIF
    Index = Index + 1
  ENDWHILE
  IF Found = TRUE THEN
    Display 'Position', Index - 1
  ELSE
    Display 'Not found'
  ENDIF
END Search
```

Question 9 (a)

Criteria	Marks
• Provides a user interface suitable for users to register an account, with features labelled	3
• Designs a user interface with some relevant features	2
• Shows some understanding of the problem	1

Sample answer:

The image shows a registration form titled "Registration Form" enclosed in a rounded rectangle. The form is divided into two main sections. The top section contains four input fields: "First name:", "Last name:", "Phone:", and "Date of birth:". The "Date of birth:" field includes a date picker icon (a small 'V' in a box). The bottom section contains three input fields: "Username:", "Password:", and "Confirm password:". The "Username:" field has an error message "Not a valid username" to its right. The "Password:" and "Confirm password:" fields are masked with asterisks. To the right of the form, several annotations with arrows point to specific features: "Phone number may have a format applied depending on country" points to the "Phone:" field; "Date picker" points to the date picker icon in the "Date of birth:" field; "Passwords are masked with '*' " points to the masked password fields; and "Error messages" points to the "Not a valid username" message. A "Submit" button is located at the bottom center of the form.

Registration Form

First name:

Last name:

Phone:

Date of birth: V

Username: Not a valid username

Password:

Confirm password:

Submit

Annotations:

- Phone number may have a format applied depending on country
- Date picker
- Passwords are masked with '*'
- Error messages

Question 9 (b)

Criteria	Marks
• Provides a correct function in Python that checks whether a username satisfies the rules	3
• Provides a partially correct function in Python that attempts to check whether a username satisfies the rules	2
• Shows some understanding of the problem and Python programming	1

Sample answer:

```
def validateUsername(username):  
  
    # Length check  
    if len(username) > 8:  
        return False  
  
    # Alpha character only check  
    elif username.isalpha() == False:  
        return False  
  
    # < character check  
    elif "<" in username:  
        return False  
  
    # If we reach here then the username meets the requirements  
    return True
```

Answers may include print statements to test the code

eg `print(validateUsername("test"))`

Question 10

Criteria	Marks
• Explains how the load time of the web page can be improved	4
• Outlines how some issues affecting the load time can be resolved	3
• Identifies issues affecting the load time OR • Outlines how to resolve ONE issue affecting the load time	2
• Provides some relevant information	1

Sample answer:

The images can be resized to 640 x 480 beforehand so that the webpage does not need to load the large 4000 x 3000 images. The multiple CSS style sheets could be converted into a single .css file, reducing the number of HTTP requests the browser needs to send. If the "image-gallery.js" JavaScript file at the top of the source code loads slowly, it can be moved to the bottom just above </body> so that it will not block the rest of the page from loading.

Answers could include:

If the "image-gallery.js" JavaScript file at the top of the source code loads slowly, the "async" property can be added to the <script> tag so that it can load in parallel with the rest of the page.

Question 11 (a)

Criteria	Marks
• Identifies the correct answer	1

Answer:

Cross site scripting

Question 11 (b)

Criteria	Marks
• Explains ONE way to minimise the vulnerability	2
• Provides some relevant information	1

Sample answer:

The website could check and change user input so that markup and script become plain text. This will still allow the page to display the information but will prevent it from executing script code.

Note:

For example, special characters such as < > and () would be encoded to their HTML equivalent (%3C %3E and %28 %29).

Question 12 (a)

Criteria	Marks
• Provides the correct output for the SQL query, including the correct ordering of the rows	2
• Provides some relevant information	1

Sample answer:

ProductName	ProductPrice
Discovery 1 & 2 (DVD)	26.98
The Plant (DVD)	28.00

Question 12 (b)

Criteria	Marks
Provides a correct SQL query	3
- Provides a substantially correct SQL query with at least TWO of the following features: - renaming the columns - selecting the three tables - joining the tables	2
Shows some understanding of renaming the columns or joining the tables	1

Sample answer:

```
SELECT Customers.CustName as 'Customer', Products.ProductName as 'Product',  
Orders.OrderQuantity as 'Quantity'  
FROM Products, Customers, Orders  
WHERE Orders.ProductID = Products.ProductID  
AND Orders.CustID = Customers.CustID  
ORDER BY Orders.OrderID ASC
```

Question 13

Criteria	Marks
- Provides a substantially correct solution including all of the following features: – looping correctly – referring to contents of the data structure correctly – displaying the player's scores and corresponding points as shown – calculating the total points – taking bonus points into account – handling the final round correctly	6
Provides a solution that shows understanding of most of the above features	5
Provides a solution that shows understanding of some features	3–4
Provides a solution that shows understanding of one feature	2
Shows some understanding of the requirement	1

Sample answer:

```
# Input code
scores = [3,5,5,2,1,4,0,1,4,5]

# Display the scores and points

# Header row print("Round\tScore\tPoints")
totalPoints = 0

# Calculate the points and print
for i in range(len(scores)):
    roundNumber = i + 1
    score = scores[i]
    points = score

    # Check if any bonuses apply and NOT the final round
    if points == 5 and i != 9:
        # Add the points from the next round as bonus points
        points += scores[i + 1]

    # Check if the final round bonus applies
    if points == 5 and i == 9:
        # Give the bonus points of 10
        points = 10

    # Add to the total points tally
    totalPoints += points

    # Print the results
    print(str(roundNumber) + "\t" + str(score) + "\t" + str(points))

print("\nTOTAL POINTS: " + str(totalPoints))
```

Question 14

Criteria	Marks
• Makes an informed judgement about the use of artificial intelligence in our society, demonstrating comprehensive understanding of its use and effects • Supports answer with information from the slideshow	8
• Shows a broad understanding of the use of artificial intelligence and its advantages and disadvantages • Includes some information from the slideshow	6–7
• Shows a sound understanding of the use of artificial intelligence • Outlines some of its advantages and/or disadvantages	4–5
• Shows some understanding of artificial intelligence	2–3
• Provides some relevant information	1

Sample answer:

- explore models of training Machine Learning (ML), including:
 - supervised learning
 - unsupervised learning
 - semi-supervised learning
 - reinforcement learning.
- investigate common applications of key Machine Learning (ML) algorithms, including:
 - data analysis and forecasting
 - virtual personal assistants
 - image recognition.
- assess the impact of automation on the individual, society and the environment, including:
 - safety of workers
 - people with disability
 - the nature and skills required for employment
 - production efficiency, waste and the environment
 - the economy and distribution of wealth.
- explore by implementation how patterns in human behaviour influence ML and AI software development, including:
 - psychological responses
 - patterns related to acute stress response
 - cultural protocols
 - belief systems.
- investigate the effect of human and data set source bias in the development of ML and AI solutions
- apply security features incorporated into software including data protection, security, privacy and regulatory compliance
- use and explain the ‘privacy by design’ approach in the development of software solutions, including:
 - proactive not reactive
 - embed privacy into design
 - respect for user privacy.

-
- evaluate the social, ethical, and legal issues and ramifications that affect people and enterprises resulting from the development and implementation of software, including:
 - employment
 - privacy
 - digital disruption.
 - data mining
 - streaming service management.

HSC Software Engineering

Familiarisation Questions

Mapping Grid

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	Secure Software Architecture – development steps	SE-12-1	2–3
2	3	Programming for the Web – SQL	SE-12-6	2–4
3	2	Programming for the Web – mail protocols	SE-12-4	3–4
4	2	Software Engineering Project – structure chart	SE-12-6	3–5
5	3	Software Engineering Project – test data	SE-12-7	2–5
6	3	Secure Software Architecture – sandboxing	SE-12-7	3–5
7 (a)	1	Software Automation – decision tree	SE-12-6	4–5
7 (b)	3	Software Automation – decision tree	SE-12-6	3–5
8	3	Software Engineering Project – algorithm	SE-12-7	3–5
9 (a)	3	Software Project – prototype and user interface (UI) design	SE-12-06	2-4
9 (b)	3	Secure Software Architecture – programming in Python	SE-12-2	3–5
10	4	Programming for the Web – load time	SE-12-8	3–6
11 (a)	1	Secure Software Architecture – XSS	SE-12-7	4–5
11 (b)	2	Secure Software Architecture – XSS	SE-12-7	4–6
12 (a)	2	Programming for the Web – SQL	SE-12-7	2–4
12 (b)	3	Programming for the Web – SQL	SE-12-7	4–6
13	6	Software Engineering Project – programming in Python	SE-12-2 SE-12-7	2–6
14	8	Software Automation – AI and ML Secure Software Architecture – privacy	SE-12-3 SE-12-5	2–6

HSC Software Engineering

Sample Examination

Total Marks: 80 | Reading Time: 10 minutes | Working Time: 2.5 hours

Section I — Multiple Choice (Questions 1–7)

Question 1 (1 mark)

Which of the following should take place during the requirements definition step when developing a software solution?

- ☐ Identifying algorithms to be used
- ☐ Documenting security vulnerabilities
- ☐ Identifying potential security threats in the software
- ☐ Documenting the needs and constraints of the software

Question 2 (1 mark)

Which of the following best describes SMTP?

- ☐ A protocol for email transmission
- ☐ A set of rules to securely transfer files across a network
- ☐ A set of rules for securing communication between computers
- ☐ A protocol used for synchronising communication across networks

Question 3 (1 mark)

A company conducts all of its consultations with its clients via its website.

Which of the following is the best way for the company to maintain the security of its clients' data?

- ☐ Implement data encryption
- ☐ Ensure that the software is user-friendly
- ☐ Focus on software that incorporates metadata
- ☐ Use software that allows for easy data integration with other platforms

Question 4 (1 mark)

Which of the following are benefits of collaborating with others when developing software solutions?

(More than one may be selected.)

- ☐ Privacy is improved.
- ☐ Security of data is increased.
- ☐ Intellectual property is protected.
- ☐ Different points of view can be considered.
- ☐ Tasks can be delegated to those with expertise.

Question 5 (1 mark)

New systems can be implemented in a number of different ways.

Match the implementation methods to their characteristics by dragging them to their corresponding boxes.

Direct	Phased	Parallel	Pilot
--------	--------	----------	-------

Implementation Method	Characteristic
	The new system runs immediately with no delay.
	The new system gradually replaces the old system.
	The new system and the old system run at the same time.
	The new system is trialled with a small group of users and then implemented entirely when ready.

Question 6 (1 mark)

A company is creating a new online application.

Which of the following is the least important consideration when choosing an open-source front-end web framework for the new application?

- ☐ Popularity of the framework on social media
- ☐ Quality of existing framework documentation
- ☐ Knowledge of the framework within the company's development team
- ☐ Compatibility of the framework with the company's existing technology

Question 7 (1 mark)

Consider the following Python code.

```
for index in range(n):  
    print(index)
```

Select the correct option to complete the equivalent pseudocode.

```
FOR index = [    ?    ]  
    Display index  
NEXT index
```

Options:

- ☐ 0 TO n
- ☐ 1 TO n
- ☐ 0 TO n-1
- ☐ 1 TO n-1

Section II — Short Answer (Questions 8–16)

Question 8 (2 marks)

The strategies listed below can be used to test the security of a web application that contains a database.

Classify each strategy as **EITHER** static application security testing (SAST) **OR** dynamic application security testing (DAST).

Strategy	Static Application Security Testing (SAST)	Dynamic Application Security Testing (DAST)
Simulate attacks on the web application	☐	☐
Check the way SQL queries are constructed	☐	☐
Test the database's web interface for vulnerabilities	☐	☐
Test the web application used to interact with the database when it is running	☐	☐
Analyse the code for connecting to the database to identify security vulnerabilities	☐	☐

Question 9 (2 marks)

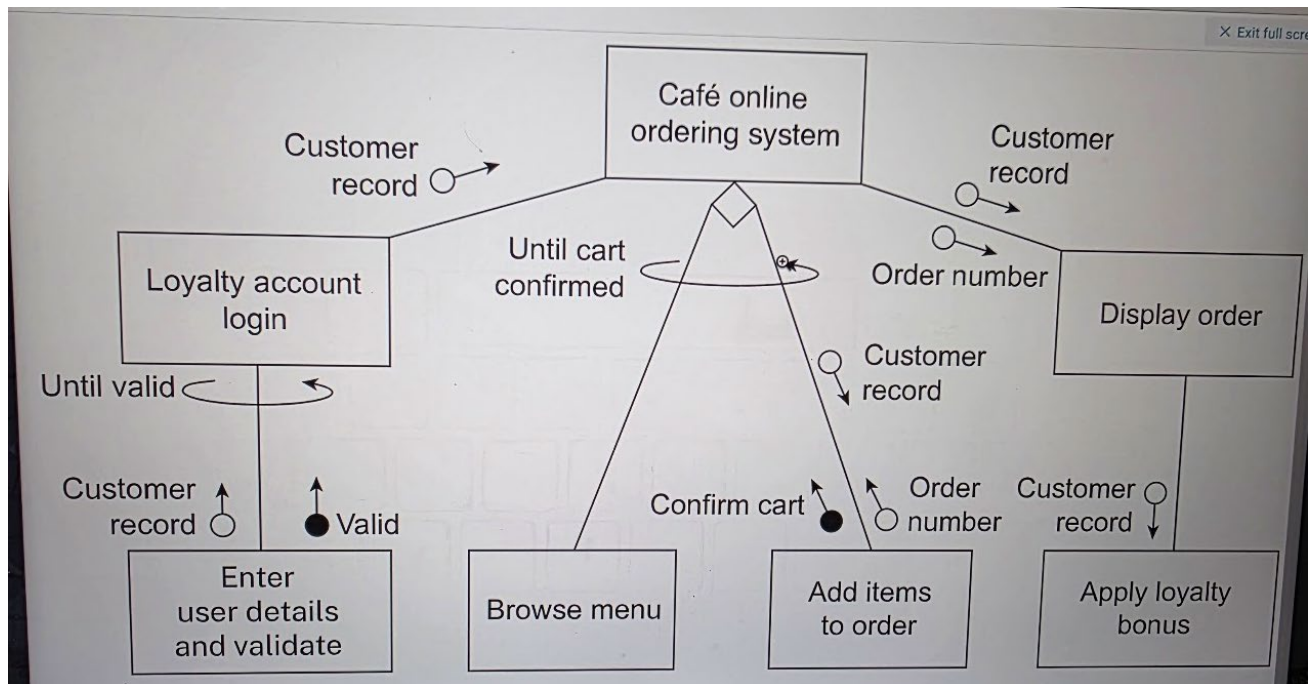
Place the following steps of a number guessing game algorithm in the correct order by dragging the steps into position.

Available Steps	Algorithm (drag into order)
Display "Guess a number between 0 and 100:"	BEGIN GuessingGame
Display "Well Done"	Set guess = -1
Set number = random integer between 0 and 100	_____
ENDWHILE	_____
WHILE guess <> number	_____
Get guess	_____

	END GuessingGame

Question 10 (2 marks)

This structure chart describes an online ordering system for a café.



Which of the following statements are TRUE about this structure chart?

(More than one may be selected.)

- ☐ **Confirm cart** is Boolean.
- ☐ Order number is a control variable.
- ☐ A valid loyalty account is not required for online ordering.
- ☐ The order cannot be displayed until the cart is confirmed.
- ☐ A choice is repeatedly offered between **Browse menu** and **Add items to order**.

Question 11 (3 marks)

Consider the following algorithm.

```
BEGIN responsiveScreen
  Get ScreenSize
  IF ScreenSize >= 1024 THEN
    Viewport = 'Desktop'
  ELSEIF ScreenSize >= 768 THEN
    Viewport = 'Tablet'
  ELSE
    Viewport = 'Phone'
  ENDIF
  Display "Display set to ", Viewport
END responsiveScreen
```

Provide a set of test data that will thoroughly test the algorithm. Include the expected outputs and reasons for inclusion. You may assume that all input data are valid.

Test data (ScreenSize)	Expected output (Viewport)	Reason for inclusion

Question 12 (6 marks)

(a) Explain how a web developer could test a website for cross-platform compatibility. **(3 marks)**

(b) Explain how the load time of a web page can be improved using a progressive web app (PWA). **(3 marks)**

Question 13 (3 marks)

Describe the function of protocols in the transfer of data. In your answer, refer to a specific web protocol.

Question 14 (3 marks)

PseudoPizza creates and sells custom-designed pizzas.

The price of a small pizza is \$10. The following costs are added, depending on the size of the pizza base and the number of toppings chosen:

- Medium size: \$2
- Large size: \$4
- Each topping: \$1.50

An algorithm is needed to calculate the cost of pizzas ordered for a customer.

The algorithm should calculate and display:

- the cost of each pizza
- the total cost of the order.

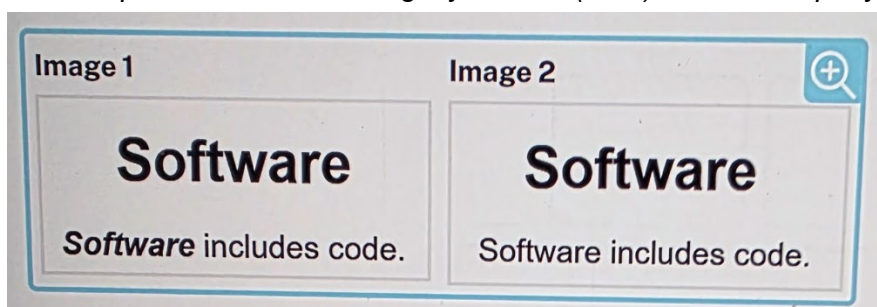
Write the algorithm using pseudocode. Include at least one subroutine in your answer. You may assume that all inputs are valid.

Question 15 (3 marks)

Describe how machine learning algorithms can be used in data analysis.

Question 16 (3 marks)

A developer added a cascading style sheet (CSS) to their company's website.



The following code was used by the developer:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Welcome to Software</title>
    <link rel="stylesheet" href="main.css">
  </head>
  <body>
    <h1>Software</h1>
    <p><span id="se_items">Software</span>
      includes code.</p>
  </body>
</html>
```

```
/* main.css */
body {
  font-family: Arial;
}

h1 {
  font-weight: bold;
  text-align: center;
}

#se_item {
  font-style: italic;
  font-weight: bold;
}
```

(a) Referring to the code, what is the cause of the problem? (1 mark)

- ☐ Font family is missing.
- ☐ Attribute of <p> has not been defined.
- ☐ main.css has not been correctly referenced.
- ☐ The id "se_item" has not been correctly referenced.

(b) Outline TWO benefits of adding CSS to the company's website. (2 marks)

Section III — Extended Response (Questions 17–25)

Question 17 (3 marks)

A shop uses an online database for recording its stock. The database contains a *Products* table and a *Suppliers* table. The prices are in dollars and the masses are in grams.

Products

ProductID	Name	Price	Mass	Producer	Quantity	SupplierID
PID001	TruffleSway	1.00	30	Yumtreats	100	SID001
PID002	PeanutPop	1.00	35	Yumtreats	500	SID002
PID003	Carameluxe	2.20	50	Yumtreats	100	SID003
PID004	BerryBlend	1.25	50	Yumtreats	8	SID001
PID005	Chocolate Cascade	1.25	52	Yumtreats	50	SID002
PID006	Hazelnut Symphony	2.50	50	Sugaryum	0	SID003
PID007	Caramel Delight	2.50	40	Treatopia	20	SID002
PID008	ChocoGlow	2.50	55	Treatopia	95	SID002

Suppliers

SupplierID	Name
SID001	The Lolly Pop
SID002	Sugar Rush Hour
SID003	Sweet Retreat

The shop wants to identify all the products with a mass of at least 50 grams that are produced by Yumtreats.

Write an SQL query to display the product names and supplier names, in descending order by product name.

Question 18 (6 marks)

An online site requires its users to create accounts. These accounts are created using a 'Create Profile' web form. This web form includes fields such as username, password, first name, last name, mobile number and date of birth.

Each username must satisfy the following rules:

1. It must be at least 8 characters long.
2. It must start with an uppercase letter.
3. The second last character must be a lowercase letter.
4. The last character must be numeric.

An example of a valid username is NES24s3.

(a) Explain how defensive data input handling practices can be implemented for this website. **(3 marks)**

(b) Write a function with Python that will check whether a username satisfies the rules. **(3 marks)**

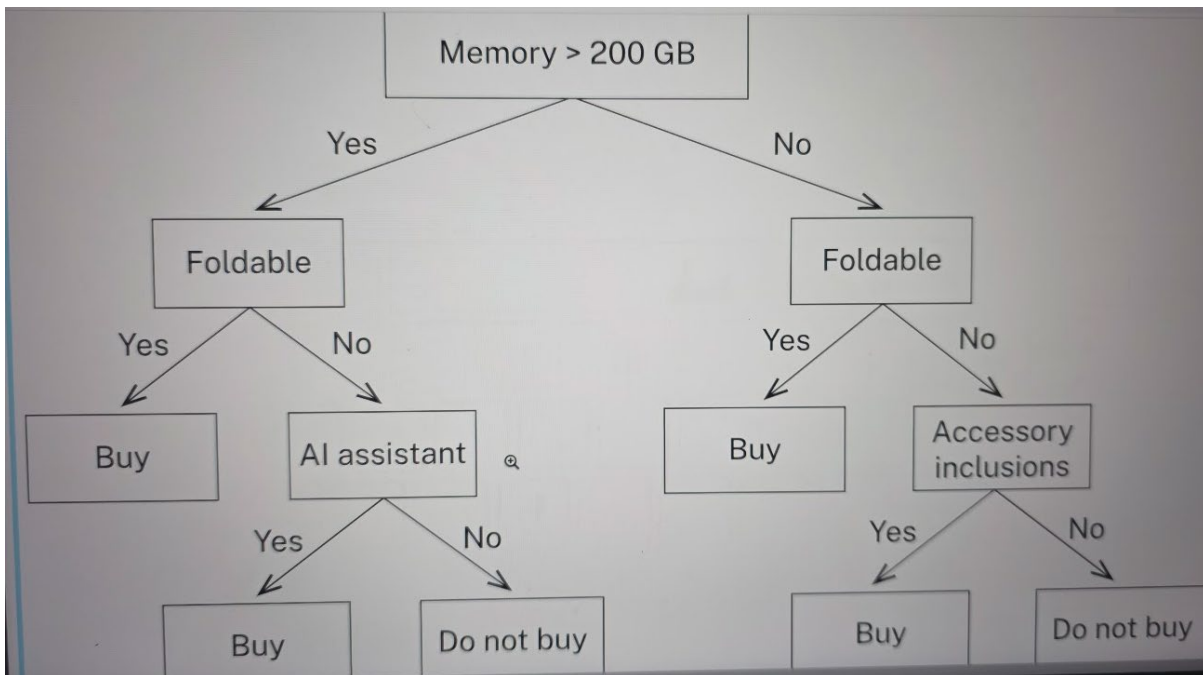
Question 19 (4 marks)

An app is being developed for use in a school. It will allow teachers to upload and assign work, and track student homework. Students can download work and upload their completed responses for marking.

Explain how authentication and authorisation could be applied to this app.

Question 20 (4 marks)

Consider the following decision tree of a trained machine learning model that determines whether to purchase a mobile phone.



(a) Using the decision tree, determine the outcome of each of the following situations. (1 mark)

Situation	Buy	Do not buy
Memory = 256 GB, AI Assistant = Yes, Accessory inclusions = No	☐	☐
Memory = 128 GB, AI Assistant = No, Accessory inclusions = Yes	☐	☐
Foldable = Yes, AI Assistant = No, Accessory inclusions = No	☐	☐

(b) The decision tree can be simplified without compromising its logic. Redraw the decision tree to reduce the number of branches. (3 marks)

Question 21 (6 marks)

At a school, each student studies five subjects. The school timetable has five periods each day, and each period is allocated to a different subject. The subjects shift to a different period each day so that each subject will appear in all five periods during the week.

An example of a student timetable is shown.

Period	Monday	Tuesday	Wednesday	Thursday	Friday
1	English	Sport	Science	Drama	Software
2	Software	English	Sport	Science	Drama
3	Drama	Software	English	Sport	Science
4	Science	Drama	Software	English	Sport
5	Sport	Science	Drama	Software	English

In this example, the student studies English, Software, Drama, Science and Sport. From Monday to Friday, English is held in Period 1, Period 2, Period 3, Period 4 and Period 5. Software is held in Period 2, Period 3, Period 4, Period 5 and then Period 1. A similar pattern can be observed for the other subjects.

You are required to write a Python program to print out the timetable in the following format:

```
Monday
  Period 1: English
  Period 2: Software
  Period 3: Drama
  Period 4: Science
  Period 5: Sport
and so on ...
```

Start your program with: `subjects = ["English", "Software", "Drama", "Science", "Sport"]`

Different students may study different subjects.

The program will be used by different students to print out their timetable by changing the subjects listed in the first line of code.

(a) If the timetable is stored as a 2D array, what is the correct Python snippet for accessing the subject that is on Monday Period 3? **(1 mark)**

- ☐ Timetable[2]
- ☐ Timetable[1,4]
- ☐ Timetable[0][2]
- ☐ Timetable[1].Subject

(b) Write the Python program using subroutines. (5 marks)

Question 22 (5 marks)

An online relational database is used to keep track of students at a coding club. The contents of the database are shown.

Students

StudentID	FirstName	Surname	Attended	Level
Student1	John	Doe	5	Beginner
Student2	Jane	Smith	4	Intermediate
Student3	David	Kim	3	Beginner
Student4	Sarah	Lee	2	Expert
Student5	Emma	Wilson	1	Beginner
Student6	Michael	Johnson	0	Intermediate
Student7	Olivia	Davis	2	Beginner
Student8	Ethan	Martinez	3	Intermediate
Student9	Sophia	Choi	4	Expert
Student10	Noah	Moore	5	Beginner
Student11	Isabella	Garcia	1	Intermediate
Student12	Liam	Vo	2	Expert

(a) Complete the SQL query to generate the number of students at each level. **Circle the correct choice in each [] (2 marks)**

```
SELECT Level, [COUNT(Level) / AVG(Level) / MAX(Level) / SUM(Level)]  
FROM Students  
[GROUP / ORDER] BY [Attended / Level / Students / Surname]
```

(b) Race conditions have been identified as a potential issue for this database. Provide an example of when a race condition may occur in this scenario and outline how secure code could be implemented to prevent it. **(3 marks)**

Question 23 (6 marks)

(a) How can the effects of human bias be minimised when training machine learning algorithms? **(3 marks)**

(b) Compare linear regression and K-nearest neighbour. **(3 marks)**

Question 24 (4 marks)

A train company wants a class diagram that models the relationship between long-distance train trips and passengers.

The long-distance train has:

- a unique trip identifier (eg AL456)
- a model code for the train engine (eg E67) and a model code (eg P9754) for each of the passenger cars
- two different areas ('Standard' and 'Executive'). The Executive area has 20 seats and there are 400 seats in the Standard area.

Each passenger:

- has contact details including an email address and phone number
- purchases a 'Standard' or 'Executive' ticket and is given a seat number (eg 25D)
- may be a frequent traveller.

Create a class diagram for the train company.



Question 25 (8 marks)

A telecommunications company had a recent security breach which prompted a review to improve the security of its systems.

It has contracted QuidantCon to propose security enhancements to its systems.

QuidantCon's Proposal — Three Slides:

Slide 1 — Business Process Automation (BPA) Security

- Predictive analysis: Identify vulnerabilities in business processes and automate risk mitigation.
- Automated compliance monitoring: Ensure continuous compliance with security regulations through ML.

Slide 2 — DevOps Integration

- Automated Security Testing: Integrate machine learning (ML) powered tools for automated code scanning and vulnerability testing.
- Anomaly Detection: Use ML to monitor logs and traffic.

Slide 3 — Robotic Process Automation (RPA) Enhancement

- Intelligent task automation: Optimise workflows with ML, ensuring compliance with security policies.
- Threat prediction: Use ML to predict and prevent security threats.

Discuss QuidantCon's proposal for security enhancements to the company's systems.

— End of Online Sample Examination —

HSC Software Engineering

Sample Examination

Marking Guidelines

Question 1

Criteria	Marks
• Identifies the correct answer	1

Answer:

Documenting the needs and constraints of the software

Question 2

Criteria	Marks
• Identifies the correct answer	1

Answer:

A protocol for email transmission

Question 3

Criteria	Marks
• Identifies the correct answer	1

Answer:

Implement data encryption

Question 4

Criteria	Marks
• Correctly identifies the benefits	1

Answer:

Different points of view can be considered.
Tasks can be delegated to those with expertise.

Question 5

Criteria	Marks
• Correctly matches the implementation methods	1

Answer:

☐ Direct	The new system runs immediately with no delay.
☐ Phased	The new system gradually replaces the old system.
☐ Parallel	The new system and the old system run at the same time.
☐ Pilot	The new system is trialled with a small group of users and then implemented entirely when ready.

Question 6

Criteria	Marks
• Identifies the correct answer	1

Answer:

Popularity of the framework on social media

Question 7

Criteria	Marks
• Identifies the correct answer	1

Answer:

```
FOR index = 0 TO n-1
    Display index
NEXT index
```

Question 8

Criteria	Marks
• Correctly classifies all the strategies	2
• Correctly classifies THREE strategies which must include ONE SAST and ONE DAST strategy	1

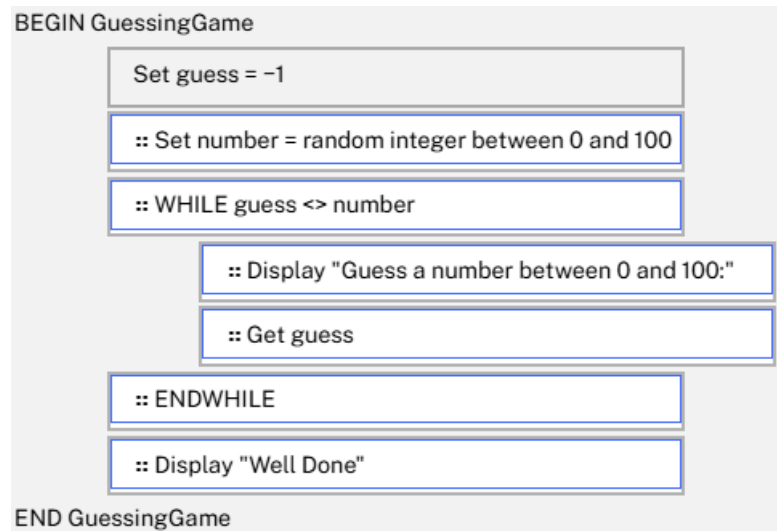
Answer:

	Static Application Security Testing (SAST)	Dynamic Application Security Testing (DAST)
Simulate attacks on the web application	☐	☒
Check the way SQL queries are constructed	☒	☐
Test the database's web interface for vulnerabilities	☐	☒
Test the web application used to interact with the database when it is running	☐	☒
Analyse the code for connecting to the database to identify security vulnerabilities	☒	☐

Question 9

Criteria	Marks
• Correctly orders the steps	2
• Correctly orders the WHILE and ENDWHILE steps	1

Answer:



Question 10

Criteria	Marks
• Correctly identifies all the items that are true about the structure chart	2
• Identifies true or false correctly for THREE out of five checkboxes	1

Answer:

- ☒ **Confirm cart** is Boolean.
- ☐ Order number is a control variable.
- ☐ A valid loyalty account is not required for online ordering.
- ☒ The order cannot be displayed until the cart is confirmed.
- ☒ A choice is repeatedly offered between **Browse menu** and **Add items to order**.

Question 11

Criteria	Marks
- Provides a set of test data that will thoroughly test the algorithm - Includes expected outputs and reasons for inclusion	3
- Provides a set of test data that tests most of the boundaries and paths - Includes the relevant expected outputs and reasons for inclusion	2
Provides some relevant information	1

Sample answer:

Test data (ScreenSize)	Expected output (Viewport)	Reason for inclusion
1024	Desktop	Boundary check
768	Tablet	Boundary check
100	Phone	Below lowest screen size check
850	Tablet	Between Tablet and Desktop
2000	Desktop	Above highest screen size check

Question 12 (a)

Criteria	Marks
Explains how a developer could test a website for cross-platform compatibility	3
- Outlines how to test a website for cross-platform compatibility OR - Identifies strategies to test a website for cross-platform compatibility	2
Provides some relevant information	1

Sample answer:

A web developer can use browser developer tools. These tools, built into most modern browsers, allow developers to simulate how their website looks and behaves on different devices and screen sizes. By using these tools, developers can identify and fix any issues related to cross-platform compatibility, ensuring that the website works well across a variety of devices and browsers.

Answers could include:

- The use of online tools built specifically for testing cross-platform compatibility
- Conducting manual testing on different devices and browsers can help identify any remaining issues
- The developer could validate the website's code using the W3C Markup Validation Service to ensure it meets the web standards.

Question 12 (b)

Criteria	Marks
Explains how the load time of a web page can be improved using a progressive web app (PWA)	3
Outlines some features of a progressive web app (PWA) and/or issues affecting load time	2
Provides some relevant information	1

Sample answer:

A progressive web app (PWA) can improve load times by pre-caching important resources, allowing files such as image-gallery.js and video-gallery.js to be loaded into memory or cache while the rest of the page loads. PWAs also use service workers to cache resources, ensuring faster load times on repeat visits and offline functionality. Background synchronisation and background fetching can update content in the background, so users see the latest information without slowing down the initial load.

Question 13

Criteria	Marks
Describes the function of protocols in the transfer of data, with reference to a specific web protocol	3
- Describes the function of protocols in the transfer of data OR - Provides an example of a web protocol and its use	2
Provides some relevant information	1

Sample answer:

Protocols are important in data transfer as they establish rules for communication between devices connected to a network, such as the internet. HTTP (Hypertext Transfer Protocol) enables browsers to request web content from servers, which respond with the requested data. The protocol defines the structure of requests and the response so that data transfer can occur.

Answers could include:

- TCP/IP
- DNS
- FTP, SFTP
- SSL, TLS
- SMTP, POP 3, IMAP
- Data packets.

Question 14

Criteria	Marks
- Provides substantially correct pseudocode - Includes at least ONE subroutine	3
Provides pseudocode that addresses some of the requirements	2
Shows some understanding of the problem	1

Sample answer:

```

BEGIN main
    Get num_pizzas
    total_cost = 0
    FOR i = 1 TO num_pizzas
        Get pizza_size

        Get num_toppings
        cost_per_pizza = calculate_pizza_cost(pizza_size,
num_toppings)
        DISPLAY cost_per_pizza
        total_cost = total_cost + cost_per_pizza
    NEXT i
    DISPLAY total_cost
END main

BEGIN calculate_pizza_cost(size, num_toppings)
    base_price = 10
    topping_price = 1.5
    IF size = "Medium" THEN
        base_price = base_price + 2
    ELSEIF size = "Large" THEN
        base_price = base_price + 4
    ENDIF
    pizza_cost = base_price + num_toppings * topping_price
    RETURN pizza_cost
END calculate_pizza_cost

```

Question 15

Criteria	Marks
• Describes how machine learning algorithms can be used in data analysis	3
• Outlines some features of machine learning algorithms relevant to data analysis	2
• Provides some relevant information	1

Sample answer:

Machine learning algorithms can be trained with a subset of the data to automatically identify patterns and trends within large datasets. The algorithms can handle complex relationships and nonlinearities present in the data allowing for efficient and comprehensive analysis. Machine learning can also be used for predictions and forecasting. These models can learn from new data over time, continuously improving their analysis and predictive accuracy and performance.

Question 16 (a)

Criteria	Marks
• Identifies the correct answer	1

Answer:

The id "se_item" has not been correctly referenced.

Question 16 (b)

Criteria	Marks
• Outlines TWO benefits of adding CSS to the company's website	2
• Provides some relevant information	1

Sample answer:

It can provide the same look and feel across the company's websites for consistency and ease of use. Separate html and CSS code can allow developers to work independently on files.

Question 17

Criteria	Marks
• Provides a correct SQL query	3
• Provides a SQL query that addresses some of the requirements	2
• Shows some understanding of SQL	1

Sample answer:

```
SELECT Products.Name AS 'Product', Suppliers.Name AS 'Supplier'
FROM Products, Suppliers
WHERE Products.SupplierID = Suppliers.SupplierID
      AND Products.Producer = 'Yumtreats'
      AND Products.Mass >= 50
ORDER BY Products.Name DESC
```

Question 18 (a)

Criteria	Marks
• Explains how defensive data input handling practices can be implemented for the website	3
• Outlines some features of data input handling practices	2
• Provides some relevant information	1

Sample answer:

Data sanitisation using client-side scripting can clean the data and remove dangerous input. Input validation on the client-side can check that entered data meets all of the requirements before getting processed by the server. Errors generated by data not meeting rules can be shown to the user.

Question 18 (b)

Criteria	Marks
• Provides a correct function in Python to check whether a username satisfies the rules	3
• Provides a partially correct function in Python to check the username	2
• Shows some understanding of the problem and Python programming	1

Sample answer:

```
def validate_username(username):
    valid = True

    # length check
    if len(username) < 8:
        valid = False
    elif (not username[0].isupper()):
        valid = False
    elif (not username[-2].islower()):
        valid = False
    elif (not username[-1].isdigit()):
        valid = False

    return valid
```

Question 19

Criteria	Marks
• Explains how authentication and authorisation could be applied to the app	4
• Explains how authentication or authorisation could be applied to the app OR • Outlines how authentication and authorisation could be applied to the app	3
• Outlines some features of authentication and/or authorisation	2
• Provides some relevant information	1

Sample answer:

Authentication ensures that only staff and students with valid usernames and passwords, which should be encrypted in storage (eg using hashing), can access the app. Authorisation controls access to specific features of the app based on user roles or groups. Teachers may have permissions to upload, assign and track homework, while students can only download tasks and upload completed work. By organising users into specific roles or groups, the app restricts access to appropriate functionality, ensuring that each user can only perform actions relevant to their role or group.

Question 20 (a)

Criteria	Marks
Correctly determines the outcomes	1

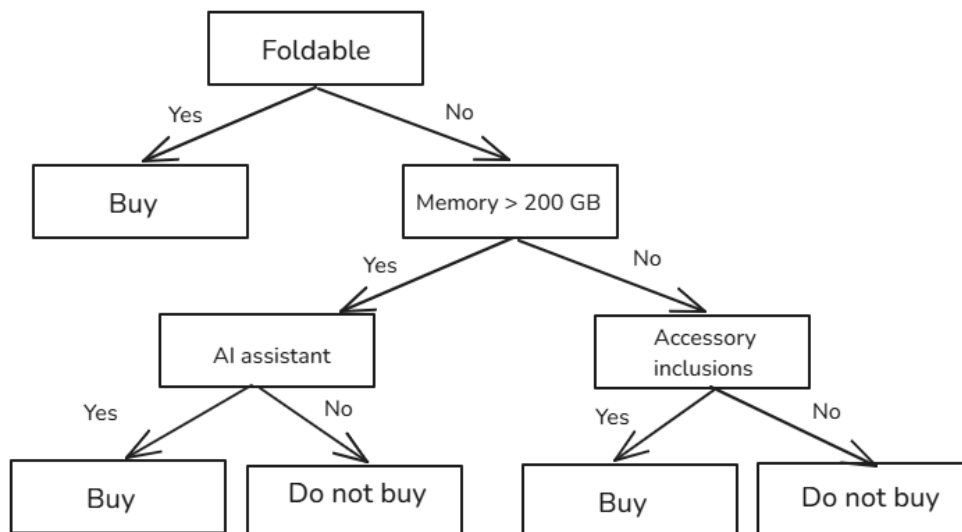
Answer:

	Buy	Do not buy
Memory = 256 GB, AI Assistant = Yes, Accessory inclusions = No	☒	☐
Memory = 128 GB, AI Assistant = No, Accessory inclusions = Yes	☒	☐
Foldable = Yes, AI Assistant = No, Accessory inclusions = No	☒	☐

Question 20 (b)

Criteria	Marks
Draws a correctly simplified decision tree	3
Draws a simplified decision tree with most of the logic	2
Shows some understanding of the requirement	1

Sample answer:



Question 21 (a)

Criteria	Marks
Identifies the correct answer	1

Answer:

Timetable[0][2]

Question 21 (b)

Criteria	Marks
- Provides a substantially correct Python program that addresses the following requirements: - Places each subject in the correct period - Displays all 5 days of the timetable - Provides a solution with appropriate subroutines - Programmatically builds the timetable	5
Provides a Python program that addresses most of the requirements	4
Provides Python code that addresses some of the requirements	3
Provides Python code that addresses a requirement	2
Shows some understanding of the problem	1

Sample answer:

```

subjects = ["English", "Software", "Drama", "Science", "Art"]
NUM_DAYS = 5
NUM_PERIODS = 5
DAYS_OF_WEEK = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday"]

def create_timetable(subjects):
    timetable = []
    for day in range(NUM_DAYS):
        periods = []
        for period in range(NUM_PERIODS):
            periods.append(subjects[(period - day) % NUM_PERIODS])
        timetable.append(periods)
    return timetable

def print_timetable(timetable):
    print()
    for day in range(NUM_DAYS):
        print(f"{DAYS_OF_WEEK[day]}")
        for period in range(NUM_PERIODS):
            print(f"\tPeriod {period+1}: {timetable[day][period]}")
        print()

timetable = create_timetable(subjects)
print_timetable(timetable)

```

Question 22 (a)

Criteria	Marks
• Correctly completes the SQL query	2
• Correctly selects at least TWO correct options	1

Answer:

SELECT Level, COUNT(Level) ▾
FROM Students
GROUP ▾ BY Level ▾

Question 22 (b)

Criteria	Marks
• Provides an example of when a race condition may occur in this scenario • Outlines how secure code could be implemented to prevent it	3
• Outlines some features of race conditions	2
• Provides some relevant information	1

Sample answer:

With multiple teachers potentially accessing the database at the same time, race conditions can allow malicious users to gain unauthorised access. Authentication could be used to restrict access to the database, and code can be written to use atomic transactions and database locks to time the reading and writing of data.

Question 23 (a)

Criteria	Marks
• Shows thorough understanding of how the effects of human bias on the training of machine learning algorithms can be minimised	3
• Outlines some effects of human bias on machine learning OR • Outlines some features of the training of machine learning algorithms	2
• Provides some relevant information	1

Sample answer:

Training data is initially selected by humans, who may unintentionally introduce their own biases. To minimise this, ensure that training data is representative of the problem the ML model will address. Reviewing data by individuals from diverse backgrounds could identify and reduce potential biases, ensuring fairer and accurate model performance.

Question 23 (b)

Criteria	Marks
• Compares linear regression and K-nearest neighbour	3
• Outlines features of linear regression and/or K-nearest neighbour	2
• Provides some relevant information	1

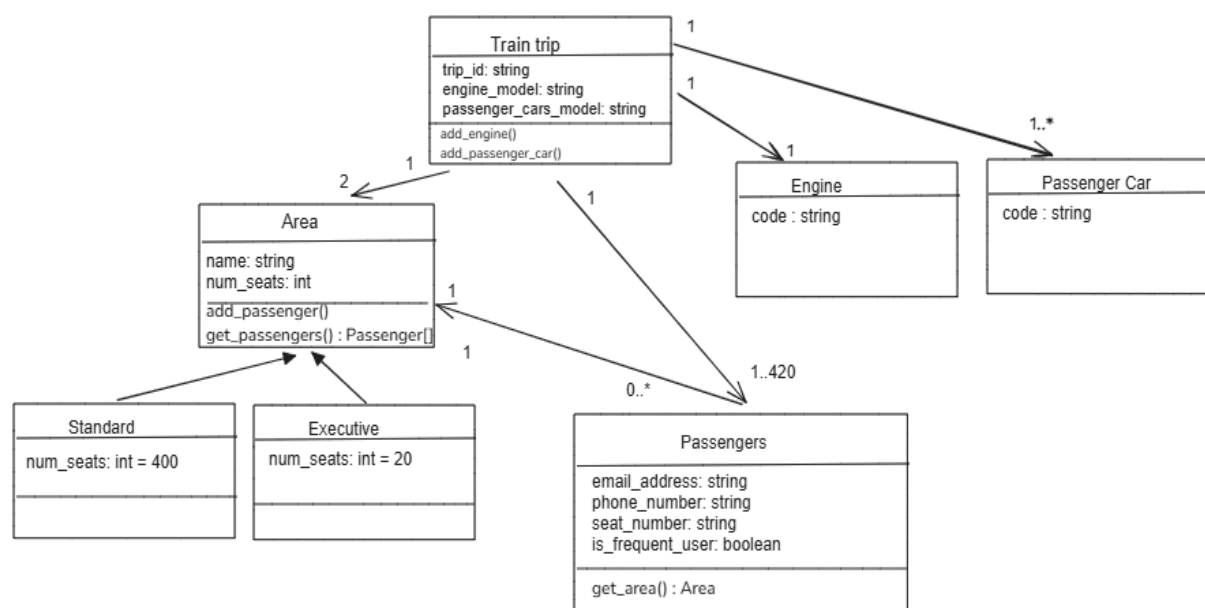
Sample answer:

Linear regression and K-nearest neighbours (KNN) are both supervised learning algorithms that make predictions from labelled data. The key difference is how they handle data: linear regression assumes a linear relationship between input features and the target variable, while KNN makes predictions based on the closest 'K' training examples, allowing it to handle more complex, non-linear patterns. However, KNN can be more computationally expensive than linear regression and is usually slower.

Question 24

Criteria	Marks
• Creates a substantially correct class diagram that includes suitable classes with attributes, and methods showing inheritance, relationships and multiplicity	4
• Creates a class diagram that represents most features of the system	3
• Creates a class diagram with some features of the system	2
• Provides a diagram that shows some understanding of the scenario	1

Sample answer:



Question 25

Criteria	Marks
Discusses the proposal for security enhancements to the company's systems, with reference to machine learning (ML) in DevOps, robotic process automation (RPA) and business process automation (BPA) from the slideshow	8
Outlines advantages and disadvantages of the proposal, with reference to at least TWO of these areas: machine learning (ML) in DevOps, robotic process automation (RPA), business process automation (BPA)	6–7
Outlines some advantages and/or disadvantages of the proposal, with reference to machine learning (ML) in DevOps and/or robotic process automation (RPA) and/or business process automation (BPA)	4–5
Shows some understanding of the features of the proposal	2–3
Provides some relevant information	1

Sample answer:

QuidantCon's proposal for the company suggests leveraging machine learning (ML) and automation to bolster security measures. One key benefit is the automation of security testing and vulnerability scanning, facilitated by integrating ML-powered tools into the DevOps pipeline. This approach ensures continuous security testing and provides rapid feedback to developers, allowing for early detection of security issues and reducing the risk of vulnerabilities being exploited. The frequent and automated testing improves the overall security position and enhances compliance with industry standards. Additionally, ML algorithms can analyse vast amounts of data to detect patterns and anomalies, which is crucial for proactive threat identification. This capability enables the company to identify security risks proactively, learn from past incidents, and predict future attacks, thereby enhancing their overall security measures.

QuidantCon's proposal for the company also raises some concerns. One key issue is the need for high-quality data for ML algorithms to work effectively. If the data is inconsistent or incomplete, the ML models may not perform well, leading to false positives or missed threats. Another concern is the complexity of implementing ML solutions, which can be challenging and may require specialised skills and training. This complexity could result in integration challenges, higher costs, and delays. Additionally, ML systems can be biased, potentially affecting security decisions. Security and privacy are also critical, as handling sensitive data requires protection against unauthorised access and breaches. These concerns emphasise the need for a careful and balanced approach when considering ML and automation in security practices.

Answers could include:

Automation of security testing and vulnerability scanning:

- Automation of security testing including the use software tools to automatically test the security of an application or system
- Conducting scans for vulnerabilities, misconfigurations, and potential threats
- Automation helps in detecting security issues early in the development lifecycle, reducing the risk of vulnerabilities being exploited
- Automation also enables more frequent testing, improving overall security posture and compliance with security standards
- Integration of automation into the DevOps pipeline, allowing for continuous security testing and rapid feedback to developers.

Use of machine learning powered tools to enhance security practices:

- Machine learning helps in identifying security risks by analysing vast amounts of data to detect patterns and anomalies that may indicate a potential threat
- Machine learning algorithms can learn from past security incidents and use this knowledge to predict and prevent future attacks
- Machine learning can automate the analysis of security logs and events, providing real time insights
- Machine learning can assist in automating incident response in RPA, enabling faster and more effective mitigation of security risks.

Safe and efficient operation of Robotic Process Automation (RPA) bots:

- Machine learning algorithms for real-time detection and response to security threats
- Machine learning algorithms can continuously monitor RPA bot behaviour to detect any anomalies or deviations from normal operation
- By analysing data on bot performance and interactions, machine learning can identify potential security threats or operational issues
- Machine learning can optimise the performance of RPA bots by identifying areas for improvement and suggesting enhancements.

Protection of Business Process Automation workflows from cyber threats

- Machine learning algorithms can analyse BPA workflows to identify potential vulnerabilities
- By continuously monitoring BPA processes, machine learning can detect and respond to cyber threats in real-time
- Machine learning can automate the implementation of security controls and policies
- Machine learning can enhance BPA workflows by predicting and preventing potential security breaches.

Additional benefits of machine learning in security:

- Improved efficiency and accuracy in detecting security threats
- Enhanced ability to respond to security incidents in real-time
- Ensuring security and compliance while automating business processes.

Disadvantages of ML in Security:

- Complexity and learning curve
- Dependency on data quality
- Potential for bias
- Security and privacy concerns
- Integration challenges
- Cost considerations.

HSC Software Engineering

2025 HSC Examination

Total Marks: 80 | Reading Time: 10 minutes | Working Time: 2.5 hours

Section I — Multiple Choice (Questions 1–11) [11 marks]

Question 1 (1 mark)

A streaming platform updated its content suggestion algorithm for users. A group of users was upgraded to the new algorithm, and feedback was collected before replacing the algorithm for all users.

Which implementation method was used?

- ☐ Pilot
- ☐ Direct
- ☐ Phased
- ☐ Parallel

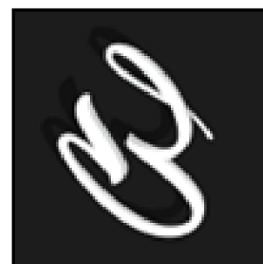
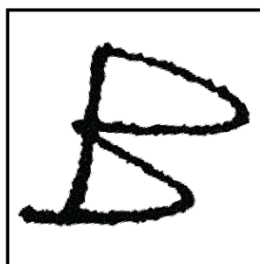
Question 2 (1 mark)

Complete the statement by selecting the correct term from each dropdown menu.

The [Agile / Waterfall / WAgile] approach typically requires extensive documentation,
while the [Agile / Waterfall / WAgile] approach often emphasises working software over documentation.

Question 3 (1 mark)

An image recognition program uses machine learning (ML) to recognise letters. It is trained using a range of images of handwritten letters. Each of the images below is labelled with the letter B when training the program.



What model of ML training is used to train this software?

- ☐ Supervised learning
- ☐ Unsupervised learning
- ☐ Reinforcement learning
- ☐ Semi-supervised learning

Question 4 (1 mark)

Fragment 1 and Fragment 2 are Python code that achieve the same purpose.

Fragment 1

```
numbers = [1, 2, 3, 4, 5]
total = 0
for num in numbers:
    total += num
print(total)
```

Fragment 2

```
numbers = [1, 2, 3, 4, 5]
print(sum(numbers))
```

Fragment 2 is an example of

- ☐ recursive code.
- ☐ list comprehensions.
- ☐ language-dependent code optimisation.
- ☐ using a hash table to create shorter solutions.

Question 5 (1 mark)

What is a potential vulnerability of an application programming interface (API)?

- ☐ Excess load times
- ☐ Biases in machine learning
- ☐ Exposure of more data than intended
- ☐ Inappropriate use of screen elements

Question 6 (1 mark)

Which of the following is a function of TLS?

- ☐ Delivering packets between hosts on the internet
- ☐ Dropping malformed packets before they reach a host computer
- ☐ Adding headers to data packets containing authentication information
- ☐ Ensuring that parties exchanging information are who they claim to be

Question 7 (1 mark)

Consider the following cascading style sheet (CSS) code fragment for a website.

```
body {  
    font-size: 100%;  
    font-style: italic;  
    line-height: 1.8;  
    font-weight: normal;  
}  
h1 {  
    text-align: center;  
    font-style: normal;  
    font-size: 250%;  
    letter-spacing: 5px;  
    background-color: rgb(0, 0, 100);  
    color: white;  
    font-weight: 900;  
    border: 3px solid green;  
}
```

Which of the following best describes the appearance of the heading on the website?

- ☐ White background, black text, green border
- ☐ Centre aligned, italic, increased letter spacing
- ☐ Blue background, white text, border around the heading
- ☐ Left aligned, bold, and 2.5 times larger than the rest of the page

Question 8 (1 mark)

File locking prevents multiple users from editing a file at the same time. When a shared file is opened by a user, file locking ensures that only this user can make changes to the file while it is locked.

Which of the following does file locking minimise?

- ☐ Race conditions
- ☐ Invalid redirection
- ☐ Cross-site scripting
- ☐ Authentication issues

Question 9 (1 mark)

Two approaches that can be used when programming databases for use in web applications are Object-Relational Mapping (ORM) and Structured Query Language (SQL).

	Object-Relational Mapping (ORM)	Structured Query Language (SQL)
Code is focused on object manipulation	☐	☐
Tables can be joined to enhance searching	☐	☐
Programmers can choose the language used to access the database	☐	☐

Question 10 (1 mark)

Which of the following are goals of regulatory compliance in relation to software development?

Select all that apply.

- ☐ Limiting the need for software updates
- ☐ Protecting sensitive data from unauthorised access
- ☐ Allowing developers to bypass security features when needed
- ☐ Reducing the testing time required during software development
- ☐ Ensuring software applications meet legal and industry standards

Question 11 (1 mark)

Which of the following are key principles of DevOps?

Select all that apply.

- ☐ Rapid feedback
- ☐ Freedom from bias
- ☐ Procedural fairness
- ☐ Automation of repetitive tasks
- ☐ Collaboration and shared ownership

Section II — Short Answer (Questions 12–20)

Question 12 (2 marks)

For which of the following situations would logistic regression be used when training a machine learning model to make predictions? Select all that apply.

- ☐ Identifying whether a received email is spam
- ☐ Determining the outcome of a two-party election
- ☐ Calculating house prices based on size of the land
- ☐ Investigating the relationship between height and weight
- ☐ Estimating the rate at which a contagious illness will spread
- ☐ Working out if someone will be admitted to a university course

Question 13 (2 marks)

A software developer plans to use static application security testing (SAST) and dynamic application security testing (DAST) to check an application that is being developed.

For each strategy, choose the most appropriate stage in software development for it to be applied. Indicate whether access to source code is required.

Testing and evaluating

Identifying and defining

Researching and planning

Producing and implementing

Yes

No

Strategy	Most appropriate stage in development	Source code access required?
SAST (Static Application Security Testing)		
DAST (Dynamic Application Security Testing)		

Question 14 (1 mark)

Consider this code fragment.

```
FOR index = 5 TO 3
    word[index] = word[index-2]
NEXT index
```

The array, word, contains the following data.

index	1	2	3	4	5
word[index]	S	M	A	R	T

Select the correct item from each dropdown menu to show the contents of the array after the code is run.

A

M

R

S

T

Question 15 (6 marks)

A computer game is being developed.

(a) Part of a data dictionary for the computer game is shown. Select the correct entries to complete the data dictionary. (3 marks)

Data type		Size in bytes	Example
Array	hmmss.sss		
Array	hh:mm:ss.sss	5	NATHAN
Boolean	hh:mm:ss	6	2ATHA2
Integer	ss:mm:hh.hhh	7	NATH22
String		8	N2222N

Variable	Data type	Format for display	Size in bytes	Description	Example
PlayerId		NXXXXN		Identifies player	
PlayerName		XX...XX	15	Player name	Nat Kim
Level		NN	1	Level of player	27
OnMission		X	1 bit	Y or N	Y
LastSave	Date and Time		12	Time last saved 24-hour time	13:45:21.221

(b) While this game is being developed, there are persistent issues with the code. Outline TWO software debugging tools that can be used to identify these logic errors. (3 marks)

Question 16 (9 marks)

A weather app is to be designed to display real-time updates such as:

- rain commencing within the next 10 minutes
- weather event warnings and warning cancellations
- location specific weather conditions.

(a) Compare the use of a progressive web app (PWA) with an interactive website, for delivering this weather app. (3 marks)

(b) Outline TWO ways, other than the use of a PWA, to minimise the load time of this weather app. **(3 marks)**

(c) Describe ONE way that the 'privacy by design' approach could be implemented in this weather app. **(3 marks)**

Question 17 (3 marks)

An algorithm is required by a school to rate the risk of a location for school excursions.

A teacher completes a 10-question survey to assess potential hazards for the location. Each question is given a score from 1 (low) to 5 (high), and the scores are stored in a 10-element integer array called Risks.

The scores for each question are added to give a total risk score. The excursion risk is rated as:

- Low (total less than 20)
- Medium (total between 20–35)
- High (total above 35).

Using pseudocode, with a repeat-until loop, write an algorithm to determine the excursion risk rating for a location. You can assume that the scores for the 10 questions are already stored in the array Risks.

--

Question 18 (3 marks)

A streaming service has exclusive rights to broadcast a live event each Sunday afternoon.

Explain how the streaming service can use metadata and data mining to prevent technical difficulties during the broadcast of this weekly event.

Question 19 (3 marks)

In an organisation, temporary passwords are automatically generated for new user accounts. Each password is created from the first three letters of the username followed by '123'.

For example, the username 'NESA' will have the password 'NES123'.

When a user logs in for the first time, they are asked for their username and password. They receive a welcome message or an error message, depending on whether the password entered follows the rules of new user accounts.

Write a Python program that asks a new user for their username and password, and then checks whether the password rule has been followed.

Question 20 (3 marks)

Describe how session management can affect the efficiency of code execution for the user.

Section III — Extended Response (Questions 21–27)

Question 21 (4 marks)

Below is the Issues table for a technical support company. FixTime, in minutes, is the time taken to complete each job.

JobID	Issue	Response	FixTime	Category	Status
0001	Cannot connect to the internet	Clear cache	22	Software	Pending
0002	Installation failure	Reinstall the printer drivers	76	Hardware	Pending
0004	Forgot password	Reset password	97	Software	Resolved
0005	Installation failure	Follow the installation guide	110	Hardware	Escalated
0009	Blue screen error	Run a system diagnostic test	120	Software	Pending
0011	Installation failure	Verify email settings	45	Software	Escalated

Construct a SQL query that displays:

- the number of pending software jobs
- the number of pending hardware jobs
- the number of escalated software jobs
- the number of escalated hardware jobs

List the results in order from the highest to lowest, based on the number of jobs.

Question 22 (3 marks)

An animal classification tool is to be developed. The classes to be used are shown below.

Mammal	Reptile
Colour	Colour
Size	Size
Birth type	Scale type
Number of teeth	Tongue type

Whale	Dog
Colour	Colour
Size	Size
Birth type	Birth type
Number of teeth	Number of teeth
Blowhole type	Breed
Migration location	Fur

Construct a class diagram that shows inheritance and removes redundant attributes for each class. Additional classes may be included if required.

You do NOT need to show the nature of the relationships between the classes.

Question 23 (9 marks)

A food delivery service offers prepackaged meals tailored to different dietary preferences, for example, vegetarian and gluten-free. It has developed a website for customers to place orders. The website is ready for testing and feedback.

The website includes:

- a menu that dynamically updates based on meal availability
- user accounts where customers can track deliveries and manage subscriptions
- a backend that processes orders and payments, updates stock and communicates with the delivery team.

(a) Outline potential social and ethical issues associated with the use of this website. **(3 marks)**

(b) Explain why server-side web programming would be needed to develop this website. Support your answer with examples from the website. **(3 marks)**

(c) Security features have been included in the design of the website. Choose the fundamental security concept that matches each of the security features. **(3 marks)**

Concept
integrity availability authorisation accountability confidentiality authentication

Feature	Concept
The system verifies that the total price displayed matches the actual total after items are added to the cart.	
The backend ensures the website remains accessible during peak order times.	
The system logs all user actions, such as changes to subscriptions or delivery addresses.	
User payment details, such as credit card numbers, are encrypted during transactions.	
A user with a basic account can place orders but cannot access admin features.	

Question 24 (5 marks)

A construction company introduces a safety monitoring system that uses artificial intelligence. Cameras and wearable sensors track worker movements and environmental conditions. The system detects unsafe behaviours through artificial intelligence, such as:

- standing too close to operating machinery
- entering restricted zones
- failure to wear personal protective equipment (PPE).

Alerts are sent in real-time to site supervisors, and long-term data is stored for further analysis.

Discuss the possible impact of this monitoring system on workers.

Question 25 (5 marks)

An airline introduces a virtual personal assistant as an initial support contact point for cancelled flights, missed connections, or lost luggage. The virtual personal assistant uses a neural network to:

- analyse support queries to direct customers to the appropriate department
- recommend appropriate tone and wording to be used by the human customer support representative
- assess data such as support turnaround time, customer ratings, and repeat contacts.

Explain how the neural network supports the virtual personal assistant. Include ONE positive and ONE negative impact of using the virtual personal assistant.

Question 26 (5 marks)

As part of a computer game, a program is needed to randomly place 15 diamonds on a 5×5 grid. Once the locations of the diamonds have been generated, the program outputs the 5×5 grid, with the location of each diamond signified by an 'X'.

An example of the output required is shown below (a 5×5 grid of '.' and 'X' characters).

```
| x |   | x |   | x |  
|   | x |   |   | x |  
|   | x | x | x |   |  
| x |   | x |   |   |  
| x | x | x | x | x |
```

Using Python, write a program that achieves this task.

Question 27 (6 marks)

Watch the video, and then answer the question.

Video link (strictly internal access only): <https://knoxnswedu.sharepoint.com/:v:/s/ss-tas/IQBSvyhE-UfjRpcTDQjL9Y9FATgYA-8ftEpbWlu8jVGcbGE?e=M15N95>

Explain how human bias could affect the usefulness of artificial intelligence (AI) solutions. Support your answer with reference to the video.

— End of 2025 Examination —

2025 HSC Software Engineering Marking Guidelines

Question 1

Criteria	Marks
• Identifies the correct answer	1

Answer:

Pilot

Question 2

Criteria	Marks
• Identifies the correct item from each dropdown	1

Answer:

Dropdown 1

Waterfall

Dropdown 2

Agile

Question 3

Criteria	Marks
• Identifies the correct answer	1

Answer:

Supervised learning

Question 4

Criteria	Marks
• Identifies the correct answer	1

Answer:

Language-dependent code optimisation

Question 5

Criteria	Marks
• Identifies the correct answer	1

Answer:

Exposure of more data than intended

Question 6

Criteria	Marks
• Identifies the correct answer	1

Answer:

Ensuring that parties exchanging information are who they claim to be

Question 7

Criteria	Marks
• Identifies the correct answer	1

Answer:

Blue background, white text, border around the heading

Question 8

Criteria	Marks
• Identifies the correct answer	1

Answer:

Race conditions

Question 9

Criteria	Marks
• Correctly matches each characteristic to the appropriate approach	1

Answer:

	Object Relational Mapping (ORM)	Structured Query Language (SQL)
Code is focused on object manipulation	●	
Tables can be joined to enhance searching		●
Programmers can choose the language used to access the database	●	

Question 10

Criteria	Marks
• Identifies ALL the items that are the goals of regulatory compliance in relation to software development	1

Answer:

- ☐ Limiting the need for software updates
- ☒ Protecting sensitive data from unauthorised access
- ☐ Allowing developers to bypass security features when needed
- ☐ Reducing the testing time required during software development
- ☒ Ensuring software applications meet legal and industry standards

Question 11

Criteria	Marks
• Identifies ALL the items that are key principles of DevOps	1

Answer:

- ☒ Rapid feedback
- ☐ Freedom from bias
- ☐ Procedural fairness
- ☒ Automation of repetitive tasks
- ☒ Collaboration and shared ownership

Question 12

Criteria	Marks
• Identifies true or false correctly for all SIX check boxes	2
• Identifies true or false correctly for FOUR of the six checkboxes	1

Answer:

- ☒ Identifying whether a received email is spam
- ☒ Determining the outcome of a two-party election
- ☐ Calculating house prices based on land size
- ☐ Investigating the relationship between height and weight
- ☐ Estimating the rate at which a contagious illness will spread
- ☒ Working out if someone will be admitted to a university course

Question 13

Criteria	Marks
• Correctly identifies the timing and source code access for each strategy	2
• Correctly identifies the timing and source code access for ONE strategy OR • Correctly identifies the timing for each strategy	1

Answer:

Strategy	Timing in software development	Access to source code
SAST	Producing and implementing ▼	Yes ▼
DAST	Testing and evaluating ▼	No ▼

Question 14

Criteria	Marks
• Identifies the correct item from each dropdown	1

Answer:

S ▼	M ▼	S ▼	M ▼	A ▼
-----	-----	-----	-----	-----

Question 15 (a)

Criteria	Marks
• Correctly completes ALL the dropdowns	3
• Correctly completes FIVE dropdowns	2
• Correctly completes THREE dropdowns	1

Answer:

Variable	Data type	Format for display	Size in bytes	Description	Example
PlayerId	String ▼	NXXXXN	6 ▼	Identifies player	2ATHA2 ▼
PlayerName	String ▼	XX...XX	15	Player name	Nat Kim
Level	Integer ▼	NN	1	Level of player	27
OnMission	Boolean ▼	X	1 bit	Y or N	Y
LastSave	Date and Time	hh:mm:ss.sss ▼	12	Time last saved 24-hour time	13:45:21.221

Question 15 (b)

Criteria	Marks
• Outlines TWO relevant software debugging tools	3
• Outlines ONE relevant software debugging tool OR • Identifies TWO relevant software debugging tools	2
• Provides some relevant information	1

Sample answer:

Breakpoints can be added to specific lines of code to halt program execution, allowing the programmer to check the value of specific variables or to step through the program line by line. Debugging output statements can be added to the code to indicate that a certain section of code has been reached or to show the contents of a variable.

Question 16 (a)

Criteria	Marks
• Compares the use of a PWA with an interactive website for the weather app	3
• Outlines features of a PWA and/or an interactive website	2
• Identifies a feature of a PWA or an interactive website	1

Sample answer:

Both PWAs and interactive websites are platform-independent, allowing users to access the app on any device with a browser. PWAs can pre-cache updates to enable offline updates when a stable internet connection is not available but may not provide real time updates for the most recent changes to weather or warnings. Meanwhile, interactive websites require a constant internet connection to provide any updates.

Answers could include:

Some PWAs prompt for initial installation or setup which might deter some users.

Question 16 (b)

Criteria	Marks
• Outlines TWO ways, other than the use of a PWA, to minimise the load time of the weather app	3
• Outlines ONE way, other than the use of a PWA, to minimise load time OR • Identifies TWO ways, other than the use of a PWA, to minimise load time	2
• Provides some relevant information	1

Sample answer:

One way would be to provide the option to select lower resolution images or text only display. Another way would be to utilise caching as this would provide high speed access to commonly requested weather data.

Question 16 (c)

Criteria	Marks
• Describes ONE way that the 'privacy by design' approach could be implemented in the weather app	3
• Outlines ONE feature of the 'privacy by design' approach	2
• Identifies a feature of the 'privacy by design' approach	1

Sample answer:

Embedded within the design of the weather app would be a process for regularly deleting identifiable data. For example, the app would likely record a person's live location to deliver relevant weather details. The privacy by design approach would incorporate a regular process to delete these locations from cache.

Question 17

Criteria	Marks
• Provides a substantially correct pseudocode algorithm	3
• Provides a pseudocode algorithm that addresses some aspects of the problem	2
• Shows some understanding of the problem or pseudocode	1

Sample answer:

BEGIN RiskRating

 Total = 0

 index = 0

 REPEAT

 Total = Total + Risks[index]

 index = index + 1

 UNTIL index = 10

 IF Total > 35 THEN

 Rating = "High"

 ELSE

 IF Total < 20 THEN

 Rating = "Low"

 ELSE

 Rating = "Medium"

 ENDIF

 ENDIF

 Display Rating

END RiskRating

OR

```
BEGIN RiskRating
    Total = 0
    index = 0
    REPEAT
        Total = Total + Risks[index]
        index = index + 1
    UNTIL index = 10
    CASEWHERE Total is
        > 35: Rating = "High"
        < 20: Rating = "Low"
        OTHERWISE: Rating = "Medium"
    END CASE
    Display Rating
END RiskRating
```

OR

```
BEGIN RiskRating
    Total = 0
    index = 0
    REPEAT
        Total = Total + Risks[index]
        index = index + 1
    UNTIL index = 10
    IF Total > 35 THEN
        Rating = "High"
    ELSEIF Total < 20 THEN
        Rating = "Low"
    ELSE
        Rating = "Medium"
    ENDIF
    Display Rating
END RiskRating
```

Question 18

Criteria	Marks
Explains how metadata and data mining can be used to prevent technical difficulties during the broadcast of the live event	3
Outlines how metadata AND/OR data mining can be used to prevent broadcast technical difficulties	2
Identifies a feature of metadata or data mining	1

Sample answer:

Metadata describes the content, structure and meaning of raw data, enabling more effective, accurate and efficient data mining. Data mining identifies user preferences and viewing patterns from large datasets, helping streaming services personalise recommendations and identify times of predicted high demand. Streaming service management uses this along with real-time user data to balance server loads and deliver smooth playback experiences.

Question 19

Criteria	Marks
Provides correct code for the problem	3
Provides partially correct code that includes a correct selection structure OR password creation	2
Shows some understanding of the scenario and Python programming	1

Sample answer:

```
uName = input("What is your username?")
password = input("What is your password?")
passwordP1 = uName[:3]
passwordP1= passwordP1 + '123'
if passwordP1 == password:
    print ('Welcome!')
else:
    print ('Incorrect password entered')
```

Question 20

Criteria	Marks
Describes how session management can affect code execution efficiency	3
Identifies some features of session management	2
Provides some relevant information	1

Sample answer:

Efficient session management limits memory consumption by restricting session data size and lifetime. Excessive reading, writing or storing large objects increases input/output operations and slows code execution for the user. Removing sessions when they are no longer required is essential to prevent unnecessary resource consumption.

Answers could include:

- Reducing database and network bottlenecks – frequent access to session stores, especially in distributed environments, can cause latency
- Preventing memory leaks – poor session handling, particularly in server-side storage, can lead to memory leaks, triggering more frequent garbage collection and degrading performance.
- Enabling scalable, responsive systems – asynchronous or lazy cleanup can further optimise execution.

Question 21

Criteria	Marks
- Provides a correct SQL query that - extracts 'Pending' and 'Escalated' jobs for the 'Software' and 'Hardware' categories - counts the jobs for each status in each category - outputs results in descending order	4
Provides an SQL query that addresses most of the requirements of the problem	3
Provides an SQL query that shows some understanding of the problem	2
Shows some understanding of SQL or the problem	1

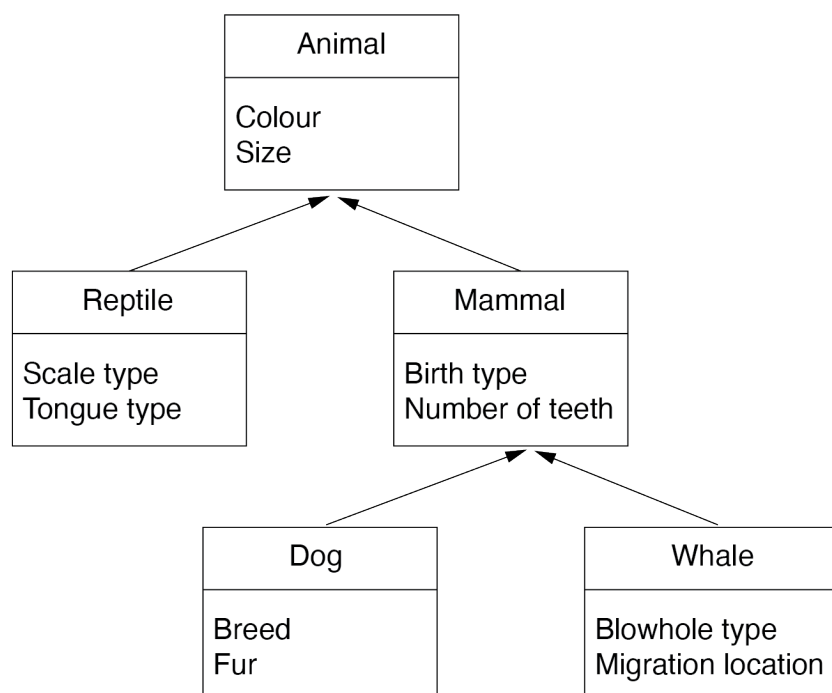
Sample answer:

```
SELECT Category, Status, COUNT(JobID) AS Jobs
FROM Issues
WHERE (Status = 'Pending' OR Status = 'Escalated')
GROUP BY Status, Category
ORDER BY Jobs DESC
```

Question 22

Criteria	Marks
Constructs a correct class diagram that shows inheritance and removes redundant attributes from each class	3
Constructs a class diagram that shows some understanding of the problem	2
Provides some relevant information	1

Sample answer:



Question 23 (a)

Criteria	Marks
Outlines relevant social and ethical issues	3
- Outlines ONE relevant social or ethical issue OR - Identifies relevant social and/or ethical issues	2
Identifies ONE relevant social or ethical issue	1

Sample answer:

The website will need to ensure payment and address data is appropriately secured and accessible only to authorised personnel. It also needs to ensure that pages can be viewed in a wide variety of languages and that the information about dishes displayed is accurate and representative of the product.

Question 23 (b)

Criteria	Marks
• Explains why server-side web programming is needed to develop the website with examples from the scenario	3
• Outlines some features of server-side web programming for developing the website with examples from the scenario	2
• Identifies a feature of server-side web programming and/or providing relevant examples from the scenario	1

Sample answer:

Server-side development will need to be used to provide dynamic information to the front-end (website). There are many dynamic aspects to the website, including the menu which changes along with subscription information and order tracking information. Server-side scripting will need to be written to read the relevant information from storage (most likely, a database) and then provide this information to the front-end.

Question 23 (c)

Criteria	Marks
• Provides the correct response for ALL items	3
• Provides the correct response for THREE items	2
• Provides the correct response for ONE item	1

Answer:

Feature	Concept
The system verifies that the total price displayed matches the actual total after items are added to the cart.	integrity ▼
The backend ensures the website remains accessible during peak order times.	availability ▼
The system logs all user actions, such as changes to subscriptions or delivery addresses.	accountability ▼
User payment details, such as credit card numbers, are encrypted during transactions.	confidentiality ▼
A user with a basic account can place orders but cannot access admin features.	authorisation ▼

Question 24

Criteria	Marks
Provides a comprehensive discussion of the impact of the safety monitoring system on workers with reference to the scenario	5
Outlines both positive and negative ways, that the safety monitoring system can affect workers	4
Outlines some ways that the safety monitoring system can affect workers	3
- Outlines ONE way that the safety monitoring system can affect workers OR - Identifies some ways that the safety monitoring system can affect workers	2
Provides some relevant information	1

Sample answer:

The AI monitoring system can improve safety on site by reducing human error through real-time alerts and remote monitoring. It can also help supervisors track workers' compliance with PPE requirements and identify risky behaviours, potentially preventing accidents. However, some workers might feel their privacy is invaded, especially if the wearables continue monitoring during breaks or in private areas like bathrooms. There's also a risk of the system misidentifying safe behaviour as unsafe, leading to unnecessary alerts or stress which could cause distractions at critical times. Additionally, the long-term data collection could be used to improve training, but workers may worry about how the data is stored or used.

Question 25

Criteria	Marks
- Explains how the neural network supports the virtual personal assistant - Includes one positive AND one negative impact of using the virtual personal assistant	5
- Explains how the neural network supports the virtual personal assistant OR - Outlines how the neural network supports the assistant and includes one positive AND one negative impact of using the virtual personal assistant	4
Outlines features of neural network AND/OR impacts of using the virtual personal assistant	3
- Outlines a feature of neural network OR an impact of using the virtual personal assistant OR - Identifies features of neural network AND/OR impacts of using the virtual personal assistant	2
Provides some relevant information	1

Sample answer:

Neural networks recognise natural language patterns to categorise common support queries, such as lost luggage, and use nodes to assist with processing. They assess language tone to gauge frustration or urgency, allowing the virtual personal assistant to recommend specific phrasing for customer service staff. For instance, calming language may be suggested for distressed customers. By tracking customer ratings and turnaround times, the neural network uses weightings to refine recommendations to prioritise high-frequency queries or improve suggestions for poorly rated interactions. While this system streamlines support processes and improves response accuracy, it may also impact the role of customer service staff. As the neural network learns from interactions, staff may rely more on its recommendations, potentially reducing their familiarity with emerging query trends or response strategies. Customers may come to expect faster, more precise service, raising the standard for human interactions.

Question 26

Criteria	Marks
- Provides a correct python program that addresses the following requirements: - creates a list of unique randomly generated numbers - processes the generated list correctly - uses selection and looping structures - provides correctly formatted output - uses inbuilt python functions	5
Provides a substantially correct python program that addresses most of the requirements	4
Provides a python program that addresses some of the requirements	3
Provides a python program that addresses one requirement	2
Shows some understanding of the problem	1

Sample answer:

```
import random
```

```
randList = []
for x in range(15):
    nextInt = random.randint(1,25)
    while nextInt in randList:
        nextInt = random.randint(1,25)
    randList.append(nextInt)
```

```
pos = 0
while pos < 25:
    if pos in randList:
        print ("| X", end=" ")
    else:
        print ("| ", end=" ")
    pos = pos + 1
    if (pos % 5 == 0):
        print("\n")
```

Question 27

Criteria	Marks
- Explains how human bias could affect the usefulness of AI solutions - Supports answer with reference to the video	6
- Describes how human bias could affect the usefulness of AI solutions - Refers to a scenario(s) in the video	5
- Outlines how human bias could affect AI solutions - Refers to the video	4
- Identifies some sources/examples of human bias AND/OR - Identifies effects of human bias on AI solutions AND/OR - Outlines how AI learns	2–3
Provides some relevant information	1

Sample answer:

AI develops its ability by analysing vast amounts of data. The quality of these data can be influenced by human bias, and in turn, affect the usefulness of AI solutions.

Human bias may be explicit or implicit. Explicit biases are intentional. For example, a researcher may favour male physicists and only collect pictures of male physicists for training an AI solution, as shown in the video. Implicit biases operate unconsciously and may be shaped by social and cultural factors. Whether the bias was explicit or implicit, the choice of male physicists would suggest to the AI solution that physicists are male.

Human bias can occur in various stages of the development of AI solutions. As shown in the photo library example, training data needs to be diverse and representative, otherwise AI solutions may produce inaccurate or unhelpful responses. The process of labelling training data may also introduce biases as people may have different interpretations of the same data. The video shows that if most people draw shoes in particular ways, certain types of shoes will not be recognised as a shoe.

2025 HSC Software Engineering Mapping Grid

Question	Marks	Content	Syllabus outcomes
1	1	Software engineering project — implementation methods	SE-12-07
2	1	Software engineering project — software development approaches	SE-12-09
3	1	Software automation — machine learning training	SE 12-03
4	1	Software engineering project — language-dependent optimisation	SE 12-08
5	1	Secure software architecture — API vulnerabilities	SE 12-04
6	1	Programming for the web — web protocols	SE 12-04
7	1	Programming for the web — CSS code fragment	SE 12-02
8	1	Secure software architecture — user action controls	SE 12-04
9	1	Programming for the web — ORM and SQL	SE 12-01
10	1	Secure software architecture — regulatory compliance	SE 12-05
11	1	Software automation — DevOps	SE 12-03
12	2	Software automation — logistic regression	SE 12-03
13	2	Secure software architecture — SAST and DAST	SE 12-04
14	1	Software engineering project — data structures	SE 12-02
15 (a)	3	Software engineering project — data dictionary	SE-12-09
15 (b)	3	Software engineering project — testing and debugging	SE 12-08
16 (a)	3	Programming for the web — load times	SE 12-06
16 (b)	3	Programming for the web — interactive websites and PWAs	SE 12-03
16 (c)	3	Secure software architecture — privacy by design	SE 12-01
17	3	Software engineering project — algorithm design	SE 12-09
18	3	Programming for the web — effect of big data on web architecture	SE 12-04
19	3	Secure software architecture — developing secure code	SE 12-07
20	3	Secure software architecture — session management	SE 12-06
21	4	Programming for the web — SQL queries	SE-12-08
22	3	Software engineering project — modelling tools	SE 12-06
23 (a)	3	Secure software architecture — social and ethical issues	SE 12-05
23 (b)	3	Programming for the web — server-side web programming	SE 12-01
23 (c)	3	Secure software architecture — fundamental software design security concepts	SE 12-04
24	5	Software automation — safety of workers	SE 12-05
25	5	Software automation — patterns in human behaviour	SE12-05
26	5	Software engineering project — design, construct and implement a solution	SE 12-07
27	6	Software automation — respond to bias in AI video stimuli	SE 12-05

Software Engineering 11–12 | Glossary of Terms

A

algorithm	A step-by-step procedure required to solve a problem. Algorithms may be presented in many ways, e.g. written instructions, flow charts or using a computer programming language.
app	A software program designed for a specific purpose to run on mobile devices or on a personal computer. An abbreviation of the word 'application'.
application programming interface (API)	An interface that allows an application or website to plug into another program or website.
artificial intelligence (AI)	Intelligence demonstrated by machines. Sometimes called machine intelligence.
assets	The digital elements that can be used or incorporated into a multimedia presentation or production. They include graphics, photographs, videos, audio, animations or other artistic data.

B

big data	Extremely large datasets that may be analysed computationally to reveal patterns, trends and associations.
black box testing	A testing technique where there is no knowledge of the internal workings of an application. A user will provide inputs and observe the outputs in order to determine functionality of the program.

C

collaborate	Working with others towards a shared goal, through a variety of modes of communication. This may be achieved using a range of technologies, tools and processes.
computational thinking	A process in which a problem is analysed and solved so that a human, machine or computer can effectively implement the solution. It involves using strategies to organise data logically, break down problems into parts, interpret patterns and design, and implement algorithms to solve problems.

D

data	A discrete representation of information using number codes. Data may include characters (e.g. alphabetic letters, numbers and symbols), images, sounds and/or instructions that, when represented by number codes, can be manipulated, stored and communicated by digital systems.
data sovereignty	A global movement that refers to the right of Indigenous Peoples to determine the creation, collection, ownership and application of data that is for and about Indigenous Peoples. The term also refers to transnational data flow and a range of issues arising from the flow of data between nations.

data types	Data types used in computing are expressed as either: string/text, character, integer, floating point or real, date and time, Boolean.
design thinking	A process where a need or opportunity is identified and a design solution is developed. The consideration of economic, environmental and social impacts that result from designed solutions are core to design thinking.
digital technologies	Refers to electronic tools, systems, devices and resources that generate, process or store data, and may include applications, games, microcontrollers, mobile devices, multimedia, networks, robotics.

E

evaluate	Make a judgement based on criteria; determine the value of.
-----------------	---

G

grey box testing	- A testing technique where there is a partial knowledge of the internal workings of an application. Its purpose is to search for defects related to improper code structure or the improper use of functions. This technique gives the ability to test both the code and the presentation layer of an application. - Note: grey box testing and gray box testing mean the same.
-------------------------	---

I

information	The presentation of data in a manner that is readily understood.
information system	The combination of digital systems, people and processes that collect, manage and analyse data.

M

model	A mathematical, conceptual or physical representation that describes, simplifies, clarifies or provides an explanation of the structure, workings or relationships within an object, system or idea. Models can provide a means of testing and predicting behaviour within limited conditions.
--------------	--

O

object-oriented programming (OOP)	A paradigm based on the concept of 'objects' that can contain data and code in the form of procedures. OOP language is a language based on the principles of 'C'.
Object-Relational Mapping (ORM)	A technique used in object-oriented programming to query and manipulate data from a database.

P

procedural programming	A method of programming where the program is divided into functions. A program consists of data and procedures (modules) that operate on the data. Data and procedures are treated as separate entities.
-------------------------------	--

prototype	A trial product or model built to test an idea or process to inform further design development. Its purpose is to see if and how well the design works and is tested by users and systems analysts. A prototype can be either a physical object or exist in digital form.
------------------	---

pseudocode	A form of algorithm description that uses English-like statements with defined rules of structure and keywords.
-------------------	---

R

refinement charts	A diagrammatic model that breaks down the steps of an outline solution into smaller and smaller steps.
--------------------------	--

S

scripting language	A programming language designed for integrating and communicating with other programming languages.
---------------------------	---

structured data	Data with a high degree of organisation, such that it can easily be analysed and processed using algorithms. The organisation of the data takes on a form that captures the properties and behaviours of the source the data represents.
------------------------	--

structured query language (SQL)	Specialist programming language used to manage data and access data in relational database management systems.
--	--

systems thinking	An understanding of how related objects or components interact to influence how systems function. Understanding the complexity of systems and the interdependence of components is important for scientific research and for the creation of solutions to technical, economic and social issues.
-------------------------	--

U

user experience (UX)	The usability, ease of use, and enjoyment provided in the interaction between the customer and the product.
-----------------------------	---

user interface (UI)	The means by which users interact with computer hardware or software. In software, this usually comprises fields for text and number entry, mouse pointers, buttons and other graphical elements.
----------------------------	---

W

W3C	The World Wide Web Consortium is the international standards organisation for the world wide web. All of the standards used for web development are defined by this organisation.
------------	---

white box testing	A testing technique used to examine the internal workings of an application including the design, code and inner workings of software. This technique gives the ability to verify the flow of inputs and outputs through the application.
--------------------------	---

NESA HSC | Glossary of Key Words

A

Account	• Account for – state reasons for, report on. • Give an account of – narrate a series of events or transactions.
Analyse	• Identify components and the relationship between them. • Draw out and relate implications.
Apply	Use in a different, new or unfamiliar situation.
Appreciate	Make a judgement about the value of.
Assess	Make a judgement of value, quality, outcomes, results or size.

C

Calculate	Ascertain/determine from given facts, figures or information.
Clarify	Make clear or plain.
Classify	Arrange or include in classes/categories.
Compare	Show how things are similar or different.
Construct	• Make. • Build. • Put together items or arguments.
Contrast	Show how things are different or opposite.
Critically analyse/evaluate	• Critically analyse: use interpretation and reasoning to assess a range of evidence and make judgements based on detailed analysis. • Critically evaluate: add a degree or level of accuracy, knowledge and understanding, logic, questioning, reflection and quality to evaluate.

D

Deduce	Draw conclusions.
Define	State meaning and identify essential qualities.
Demonstrate	Show by example.
Describe	Provide characteristics and features.
Discuss	Identify issues and provide points for and/or against.

Distinguish

- Recognise or note/indicate as being distinct or different from.
- To note differences between.

E**Evaluate**

- Make a judgement based on criteria.
- Determine the value of.

Examine

Inquire into.

Explain

- Relate cause and effect.
- Make the relationships between things evident.
- Provide why and/or how.

Extract

Choose relevant and/or appropriate details.

Extrapolate

Infer from what is known.

I**Identify**

Recognise and name.

Interpret

Draw meaning from.

Investigate

Plan, inquire into and draw conclusions about.

J**Justify**

Support an argument or conclusion.

O**Outline**

Sketch in general terms; indicate the main features of.

P**Predict**

Suggest what may happen based on available information.

Propose

Put forward (for example a point of view, idea, argument, suggestion) for consideration or action.

R**Recall**

Present remembered ideas, facts or experiences.

Recommend

Provide reasons in favour.

Recount

Retell a series of events.

S**Summarise**

Express, concisely, the relevant details.

Synthesise

Putting together various elements to make a whole.

Source: NSW Education Standards Authority (NESA) | nsw.gov.au/education-and-training/nesa/hsc/student-guide/glossary

Higher School Certificate Course Specifications

Software Engineering

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Introduction

Software Engineering Course Specifications are an integral part of the course content for Year 11 and Year 12 and indicate the depth of study required for some concepts in the *Software Engineering 11–12 Syllabus*. The *Software Engineering 11–12 Syllabus* must be applied in conjunction with the Software Engineering Course Specifications.

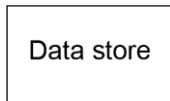
System and Data Modelling Tools

Data flow diagrams

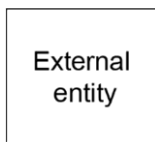
Symbols



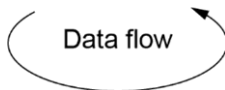
A circle represents a process. A process uses input(s) to generate output(s).



A data store can be an electronic file or non-computer storage.

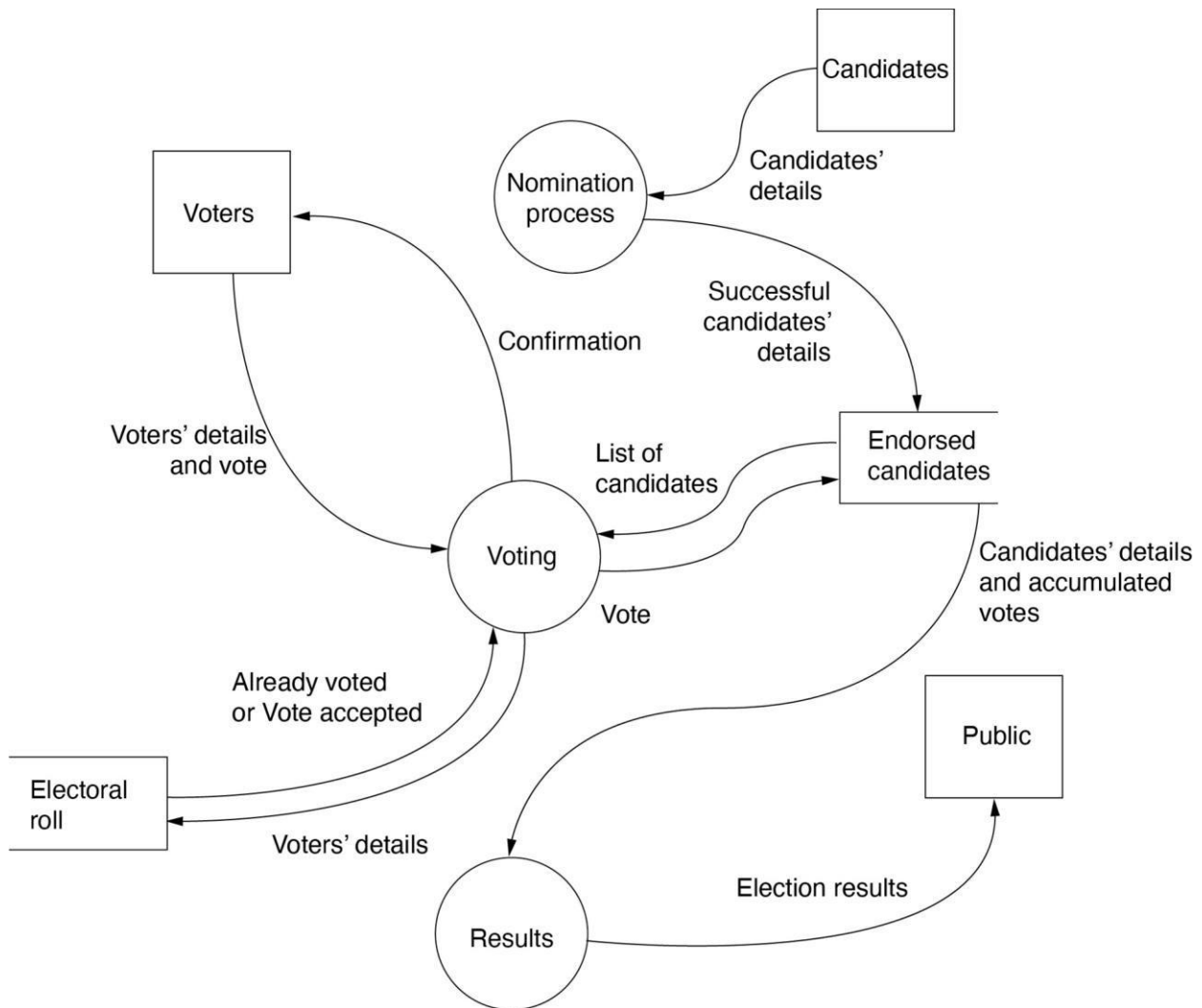


An external entity can be any person, organisation or element that provides data to the system or receives data from the system.



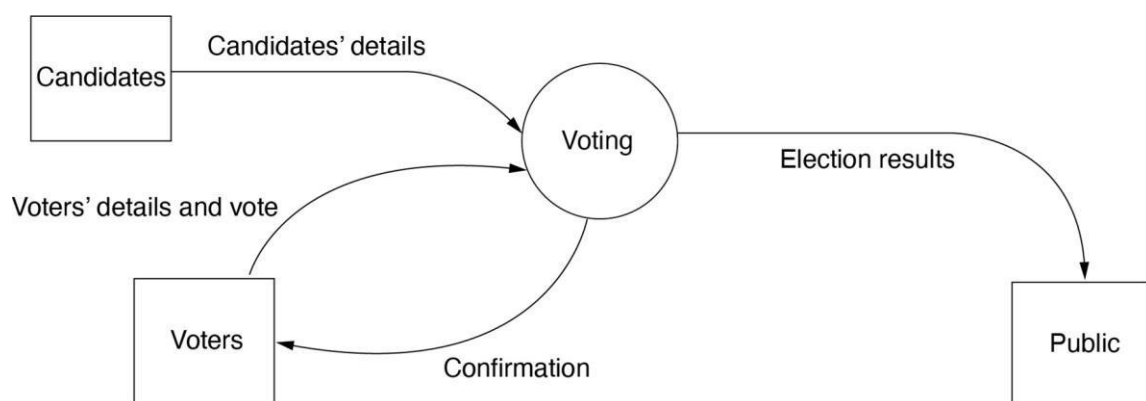
A labelled, curved arrow represents the flow of data between processes, data stores and external entities.

The following data flow diagram models a voting system.



Level 0 data flow diagram

Level 0 data flow diagrams represent an overview of the entire system and do not show data stores or internal processes. The following represents a Level 0 data flow diagram for the voting system.



Structure charts

Structure charts represent a system by showing the separate subroutines that make up the system and their relationship to each other.

Symbols



An empty circle is used to indicate data movement between subroutines (usually passed as parameters).



A filled circle is used to indicate a flag or control variable that is passed between subroutines.

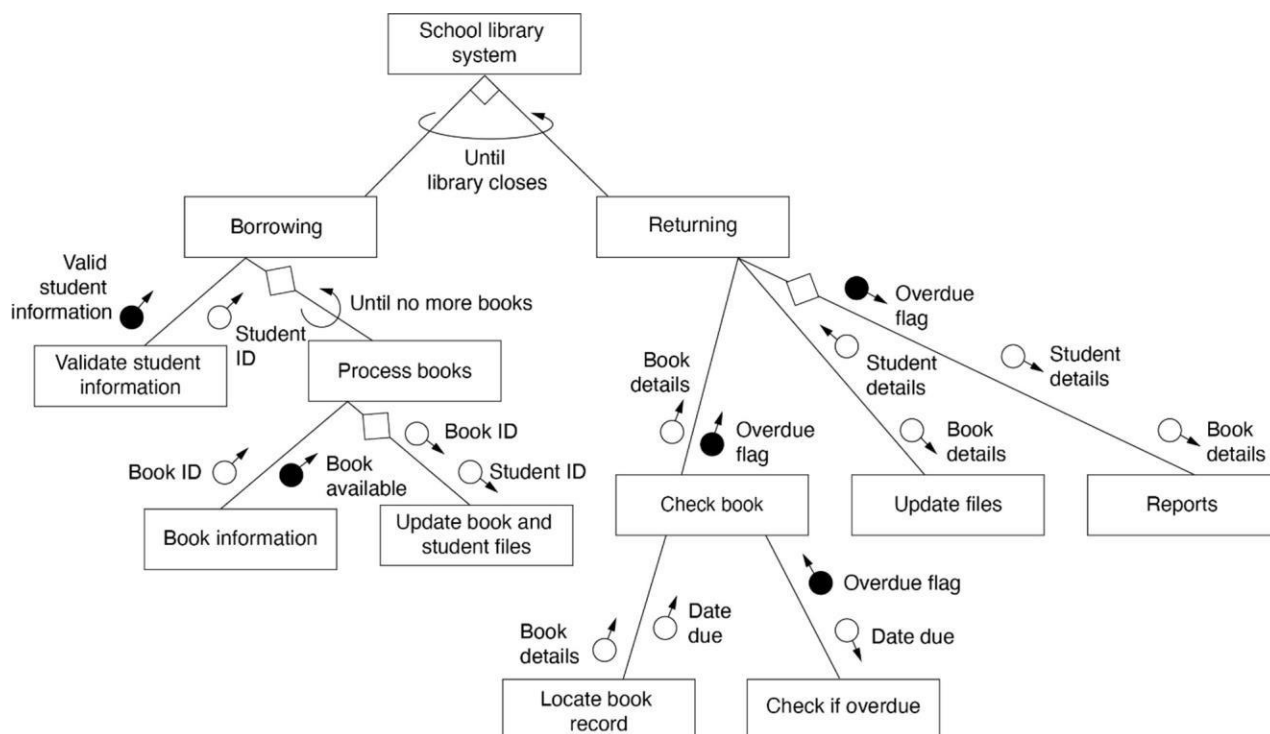


A decision (ie optional execution of a subroutine) is indicated by use of a small diamond at the intersection of the connecting lines between subroutines that are called as the result of a binary or multi-way selection. Alternatively, the diamond may appear on a single connecting line if calling that subroutine is optional. In the diagram shown, a report may not necessarily be produced each time a book is returned.



Repetition (execution of a particular subroutine or set of subroutines multiple times) is shown by a curved arrow.

The following structure chart represents a library system.



Further detail for each of the lower-level subroutines can be shown in a separate structure chart, using the same name as the subroutine used in the main structure chart.

This method of providing successively more detail as required is known as *refinement*.

Data dictionary

A data dictionary provides a comprehensive description of each variable stored or referred to in a system. This commonly includes variable name, data type, format, size in bytes, number of characters to display the item including number of decimal places (if applicable), the purpose of each variable and a relevant example. Any validation rules applicable to the data item can also be included.

Details of records or arrays of records can be included in data dictionaries.

An example of a data dictionary is shown.

Variable	Data type	Format for display	Size in bytes	Size for display	Description	Example	Validation
UserId	String	XXNNN	5	5	A primary key, uniquely identifies user. First 2 letters of surname followed by unique 3-digit identifier	PT173	First 2 characters letters, followed by last 3 characters digits
UserName	String	XX..XX	15	15	Username of employee	Kim	First letter is a capital letter
DOB	Date and Time	YYYY/MM/DD	4	10	Birth date of employee	1953/10/05	Valid date less than today
Times_Late	Integer	NNN	2	3	Number of times late to work this year	147	Integer between 0 and 999
PayRate	Floating Point	\$NNN.NN	4	7	Hourly rate of pay	\$124.37	Decimal greater than 20, less than 400
SocialClub	Boolean	X	1 bit	1	Y or N	N	
Departments	Array (string)		20 * number of departments	N/A	Names of departments in organisation	Administration Finance Marketing	From a drop-down list

Note: a date and time data type is always stored as 32 bits (4 bytes) and can be displayed using different formats such as DD/MM/YYYY hh:mm:ss or YYYY/MM/DD

Class diagrams

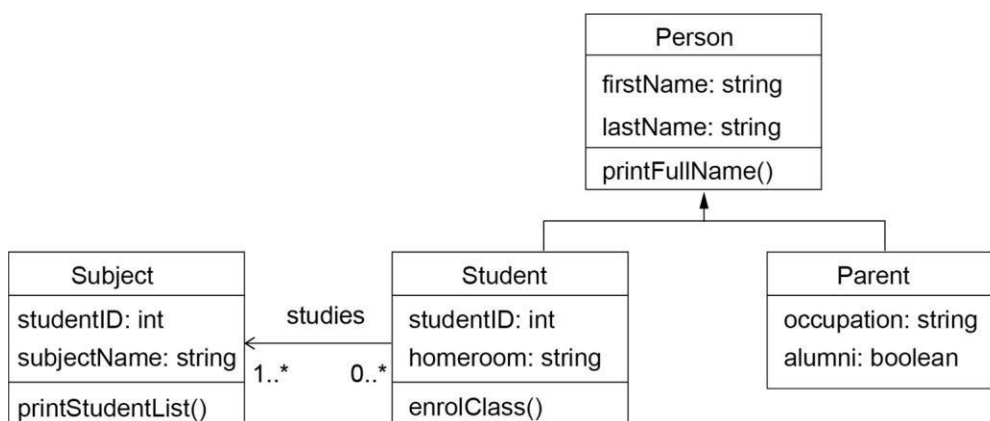
Class diagrams provide a visual representation of systems that are implemented using the object-oriented paradigm. They model classes, their attributes and methods, and the relationships between classes.

Symbols

class name	
attribute(s)	
method(s)	

	inheritance
	relationship
1..1	only one
0..*	zero or more
1..*	one or more
0..1	zero or only one
m..n	at least m, at most n ($m \leq n$)

The following is an example of a class diagram.

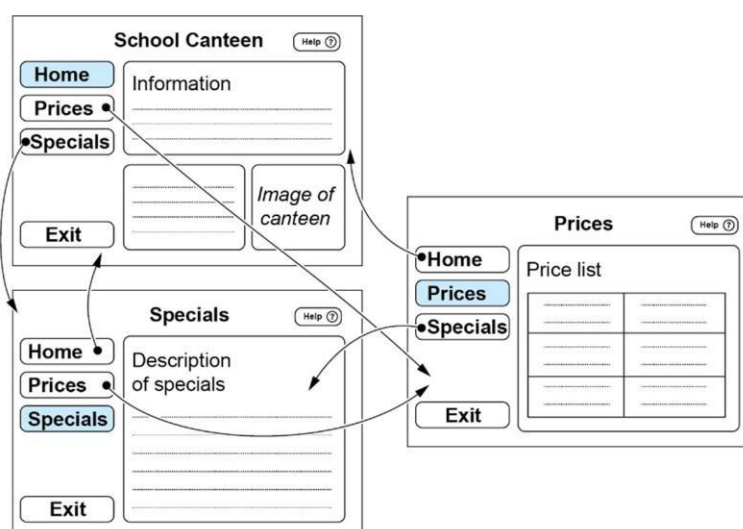


Note: Both **Student** and **Parent** inherit from **Person**. There is a relationship between **Student** and **Subject**. A student must study 1 or more subjects. A subject can have 0 or more students enrolled.

Storyboard

A storyboard shows the various interfaces (screens) as well as the links between them.

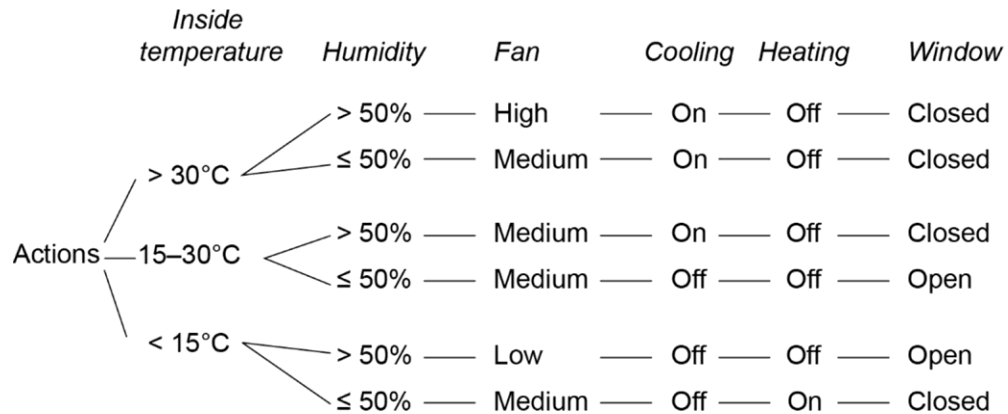
The following storyboard shows the relationship between three pages of information aimed at promoting a school canteen on a website.



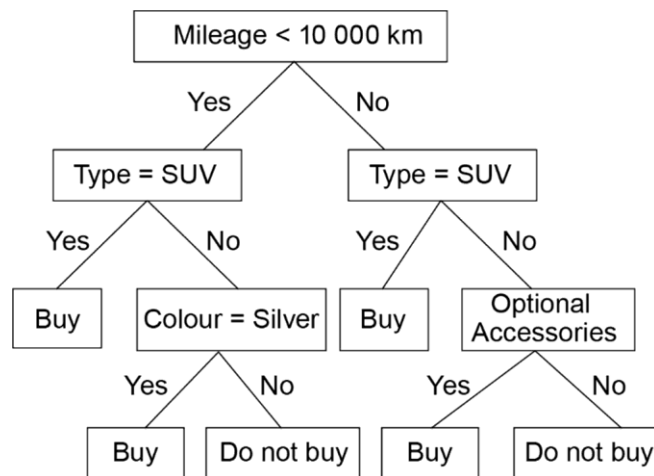
Decision trees

A decision tree is a diagram that represents all possible combinations of decisions and their resulting actions. Branches are shown to describe the eventual action depending on the condition at the time. Each decision path will lead to either another decision or a final action.

The following decision tree shows the rules in controlling the temperature system within a 'smart' house.



The following diagram shows another way to represent a decision tree.

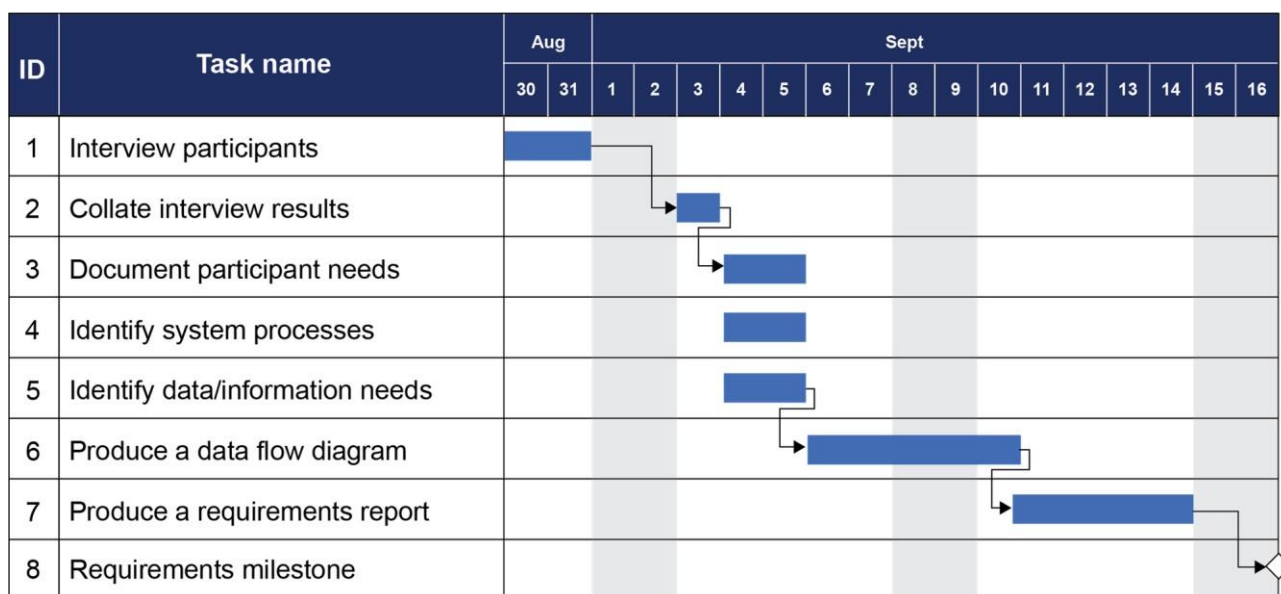


Project Management Tools

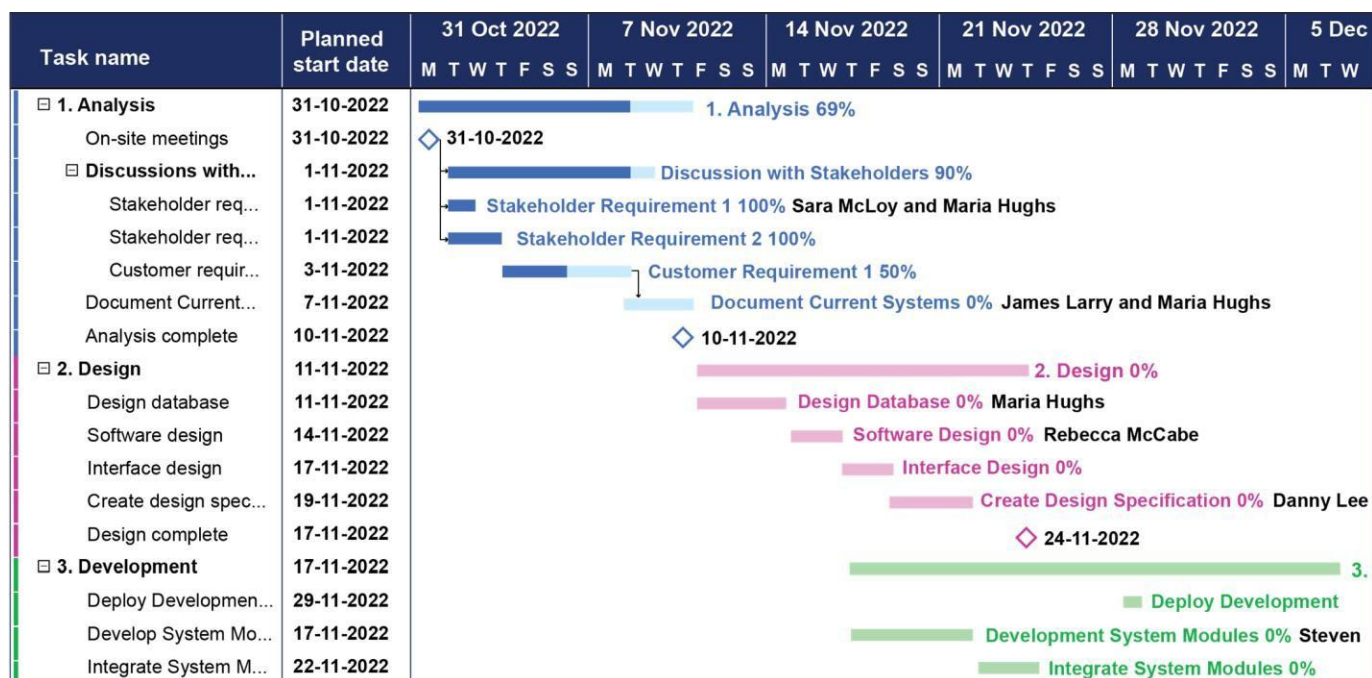
Gantt charts

A Gantt chart displays each of the component tasks in a proposed system development on an estimated timeline. Tasks should be named with self-explanatory titles. The estimated time required for each task and its dependent tasks should be clearly shown. The time scale should be clearly indicated with dates and important milestones in the project clearly marked.

The following diagram shows the main elements of a Gantt chart. Other formats are acceptable.



Gantt charts can also be used to allocate resources, including team members, to specific tasks. The following chart shows the percentage completion of tasks by each team member. Charts should be regularly updated during development to reflect actual versus estimated times for tasks.



Process diaries / log books

Process diaries / log books are used to document the progress of a project. Entries made by team members at regular intervals should include:

- date
- person making the entry
- progress since the last entry
- tasks achieved
- stumbling blocks or issues encountered and how they were managed
- possible approaches for upcoming tasks
- reflective comments
- resources used.

Programming Paradigms

Object-oriented paradigm

Students should know how to:

- define classes, objects, attributes and methods
- make use of inheritance, polymorphism and encapsulation
- perform message passing
- use control structures and variables.

Logic paradigm

Students should know how to:

- define and edit facts
- create, edit and remove rules
- display the solution and the rules that the system used.

Imperative paradigm

Students should know how to:

- use control structures and variables
- use assignment statements
- use expressions
- use subroutines.

Functional paradigm

Students should know how to:

- call functions and use recursion
- use functions as first-class objects and collections
- use abstraction, encapsulation, inheritance and polymorphism.

Note: Students should know how to use appropriate data structures for each of the paradigms.

Algorithms

It is expected that students are able to develop and interpret algorithms represented as pseudocode and flowcharts.

It is important to start complex algorithms with a clear, uncluttered mainline. The mainline should reference required subroutines, the details of which are shown in separate algorithms.

Each subroutine should be concise and correctly make use of further subroutines for detailed logic.

Pseudocode

Pseudocode is a method of describing the logic in an algorithm. It makes use of capitalised keywords and indentation to show control structures used.

In pseudocode:

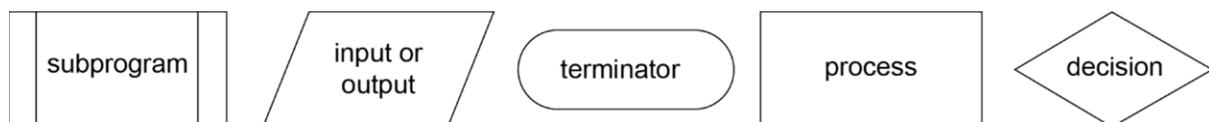
- keywords are written in capitals
- structural elements come in pairs, eg for every BEGIN there is an END, for every IF there is an ENDIF
- indenting is used to identify control structures in the algorithm
- when refining the solution to a problem, a subroutine can be referred to in an algorithm by its name, with a separate subroutine developed with that same name to show the detailed logic.

Flowcharts

Flowcharts are diagrams that represent algorithms and are read from top to bottom and left to right.

Symbols

Flowcharts use the following symbols connected by lines with arrowheads to indicate the flow of data. It is common practice to show arrowheads to avoid ambiguity.



Flowcharts using these symbols should be developed using only the standard control structures (as described in the following section, Control Structures).

Control Structures

Algorithms are developed using the basic control structures of *sequence*, *selection* and *repetition*. Students are expected to design algorithms and write code incorporating combinations of these control structures.

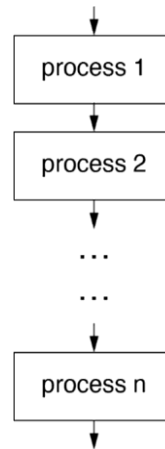
Sequence

Sequence refers to steps which are to be executed one after the other. The steps are executed in the same order in which they are written.

Pseudocode

```
process 1  
process 2  
...  
...  
process n
```

Flowchart



Selection

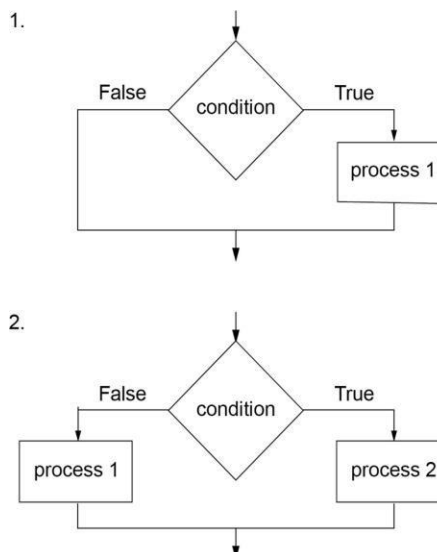
Binary selection

In binary selection, if the condition is met then one path is taken, otherwise the second possible path is followed.

Pseudocode

```
1. IF condition THEN  
    process 1  
ENDIF  
  
2. IF condition THEN  
    process 2  
ELSE  
    process 1  
ENDIF
```

Flowchart



Note: Arrows coming from a decision symbol should be labelled to remove ambiguity.

Multi-way selection

In multi-way selection there can be a number of possible choices, or cases. The path taken is determined by the evaluation of the expression. Once a relevant path has been determined and executed, execution of this expression ceases. Only one process is executed as a result of the implementation of the multi-way selection.

Multi-way selection is often referred to as a *case structure*.

Pseudocode

CASEWHERE expression evaluates to

choice a: process a

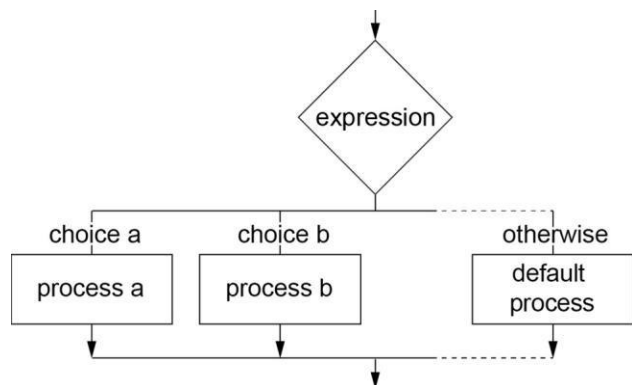
choice b: process b

...

OTHERWISE: default process

END CASE

Flowchart



Nested IF

A nested IF allows the testing of multiple conditions with only one process executed.

Pseudocode

IF condition A THEN

process 1

ELSE

IF condition B THEN

process 2

ELSE

IF condition C THEN

process 3

ELSE

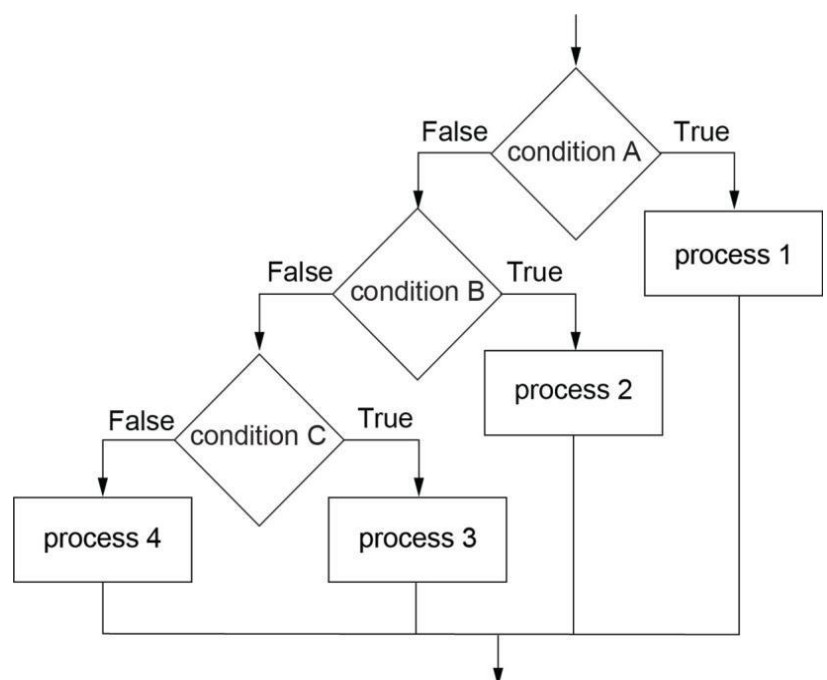
process 4

ENDIF

ENDIF

ENDIF

Flowchart



Repetition

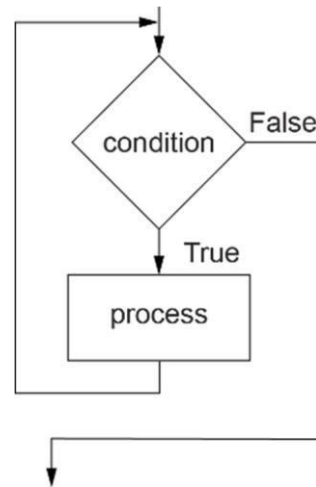
Pre-test

The pre-test loop tests the condition at the start of the loop to determine whether the body of the loop is executed. The body of the loop is executed repeatedly while the termination condition is true.

Pseudocode

```
WHILE condition is true  
    process  
ENDWHILE
```

Flowchart



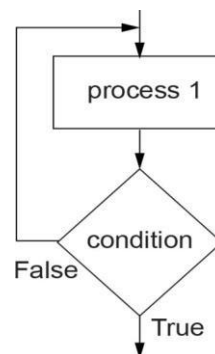
Post-test

A post-test loop executes the body of the loop before testing the termination condition. The body of the loop is repeatedly executed until the termination condition is true.

Pseudocode

```
REPEAT  
    process  
UNTIL condition is true
```

Flowchart



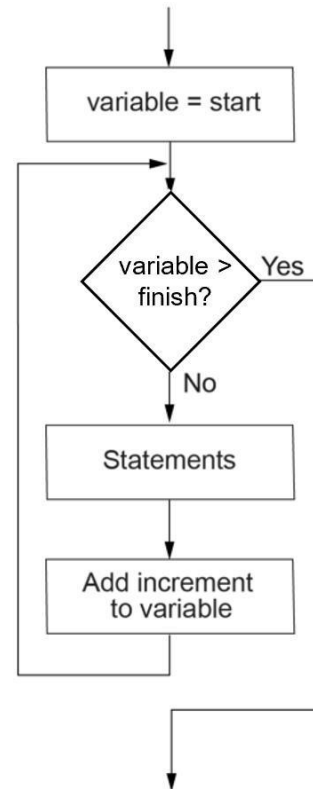
FOR / NEXT

FOR / NEXT loops (also known as counted loops) can be regarded as special cases of repetition and, depending on the language, are implemented as either pre-test or post-test repetitions.

Pseudocode

```
FOR variable = start TO finish STEP increment  
    statements  
NEXT variable
```

Flowchart



Note: Increment can take either a positive or negative value. The flowchart shows the logic for a positive increment

Subroutines

The terms *subroutine*, *module*, *subprogram* and *procedure* can be used interchangeably to represent a collection of statements that achieve a specific purpose.

A subroutine can do the same task at different points in an algorithm. It may operate on different data each time it is called. One or more parameters are used to indicate the data to be processed.

A *function* is a particular type of subroutine that returns a single value.

Using a subroutine with one parameter

The following algorithms represent the logic required to fill an array with characters.

The subroutine *read* uses a single parameter *arrayname* that can take different values.

The first time that this subroutine is called, the data is read and stored in the array called 'name'. The second time, the data is read and stored in the array called 'address'.

Pseudocode

BEGIN

 read (name)

 read (address)

END

BEGIN read (arrayname)

 Set pointer to first position

 Get a character

 WHILE more data AND space in array

 Store data in arrayname at the
 position given by the pointer

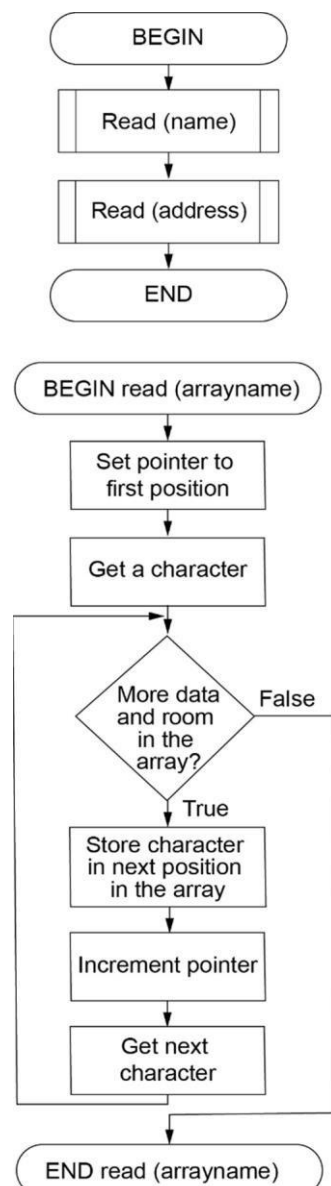
 Increment the pointer

 Get next character

 ENDWHILE

END read (arrayname)

Flowcharts



Using a subroutine with multiple parameters

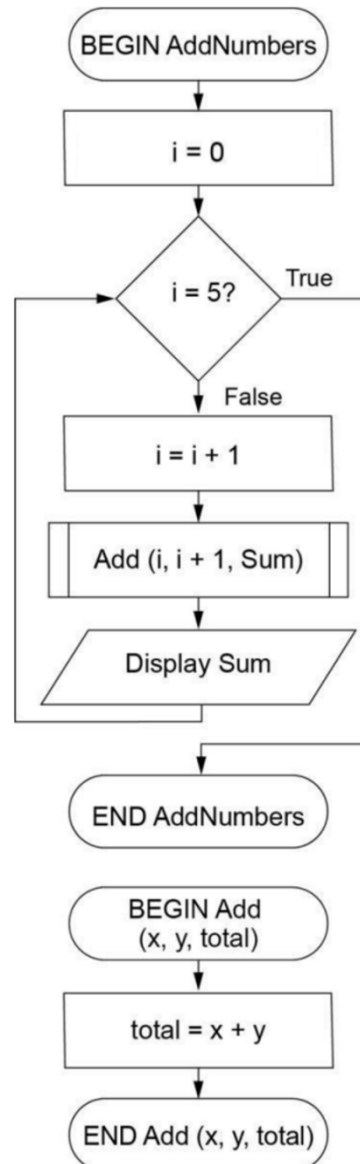
The following algorithms represent the logic required to display the sum of two consecutive integers, where the smaller integer takes all possible values from 1 to 5. The subroutine *Add* uses three parameters, *x*, *y* and *total*.

Pseudocode

```
BEGIN AddNumbers  
    FOR i = 1 to 5  
        Add (i, i + 1, sum)  
        Display sum  
    NEXT i  
END AddNumbers
```

```
BEGIN Add (x, y, total)  
    total = x + y  
END Add (x, y, total)
```

Flowcharts



Passing a value back from a function

A function generates a single value. The word RETURN is used to pass this single value back from the function.

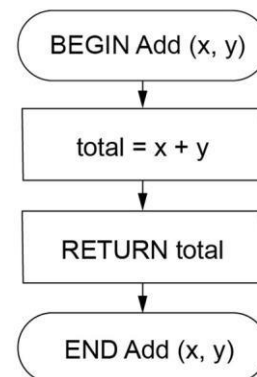
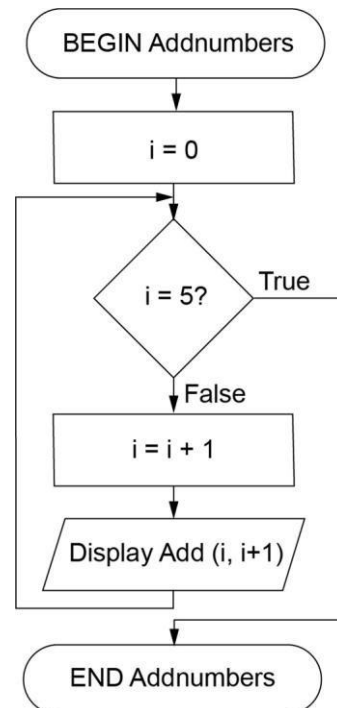
The algorithm *Addnumbers* uses the function *Add* to calculate the sum of two consecutive integers where the smaller integer takes all possible values from 1 to 5. In this case the function requires two parameters.

Pseudocode

```
BEGIN Addnumbers  
    FOR i = 1 to 5  
        Display Add (i, i + 1)  
    NEXT i  
END Addnumbers
```

```
BEGIN Add (x, y)  
    total = x + y  
    RETURN total  
END Add (x, y)
```

Flowcharts



Relational Databases

SQL

Structured Query Language (SQL) is a language used to access and manipulate data in relational databases.

For the HSC Software Engineering course, the following syntax is used.

```
SELECT the field(s) or calculated values to be displayed
FROM   the table(s) to be used
WHERE  the search criteria
GROUP BY the field(s) used to group the returned rows
ORDER BY the field(s) that determine the sequence of the displayed results
```

The keyword AS may be used within a SELECT statement to rename fields for display.

The search criteria may use relational operators, including the following.

```
CONTAINS
DOES NOT CONTAIN
EQUALS
NOT EQUAL TO
GREATER THAN
GREATER THAN OR EQUAL TO
LESS THAN
LESS THAN OR EQUAL TO
```

Logical operators that may be used include the following.

```
AND
OR
NOT
```

If the GROUP BY clause is used, it is useful to include in the SELECT statement details of the value to be calculated. The following functions may be used.

```
SUM (attribute)
AVG (attribute)
COUNT (attribute)
MAX (attribute)
MIN (attribute)
```

If the ORDER BY clause is used, the field and method should be specified. The methods used are typically identified by ASC (ascending: A – Z or 0 – 9) or DESC (descending: Z – A or 9 – 0).

Three tables from a relational database are shown.



The following query displays the name and release date of all games released from 1 March 2022 to 31 March 2023. The results will be displayed in ascending alphabetical order by game name.

```

SELECT Name, Release_date
FROM Games
WHERE Release_date >= '01/03/2022' AND Release_date <= '31/03/2023'
ORDER BY Name ASC
  
```

The following query displays each developer, together with the total cost of games they have developed for the publisher 'Games Inc', listed in descending order of developer name.

```

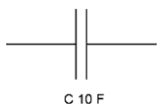
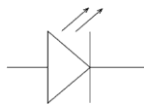
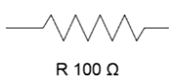
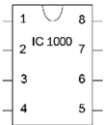
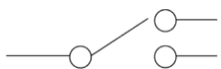
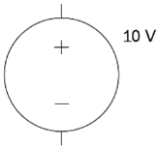
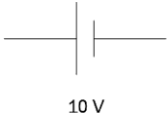
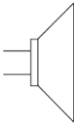
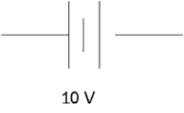
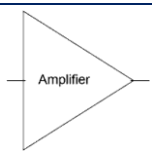
SELECT Developers.First_name, Developers.Last_name, SUM (Games.cost) AS Totalcost
FROM Games, Publishers, Developers
WHERE Publishers.Name = 'Games Inc'
AND Publishers.Publisher_ID = Games.Publisher_ID
AND Developers.Developer_ID = Games.Developer_ID
GROUP BY Developers.Developer_ID
ORDER BY Developers.Last_name DESC
  
```

Object-Relational Mapping (ORM)

ORM (Object-Relational Mapping) provides a layer of abstraction between the database and the programming language which the developer is using. In an object-oriented language, an ORM will usually be able to represent database items (rows) as objects in that chosen language and the attributes of that item (columns) as properties of the object. Students are expected to interpret code fragments used in an ORM framework.

Wiring diagrams for mechatronic systems

Mechatronic systems can range in complexity and can be represented using the following symbols.

Name	Symbol	Name	Symbol
Capacitor	 C 10 F	LED	
Diode		Lightbulb	
Resistor	 R 100 Ω	Integrated circuit	
2-way Switch		Voltage source	 10 V
On/off Switch		DC Voltage Source	 10 V
Speaker		DC Voltage source	 10 V
Motor		Amplifier	 Amplifier

If a wiring diagram requires other electrical components, the components must be clearly labelled for identification.

Programming for the Web

Front-end web development frameworks

There are numerous front-end web development frameworks which provide different features and benefits. Students should understand why such frameworks are useful in front-end web development

Students are not expected to have knowledge of a specific framework nor are they expected to code using any specific framework.

Cross-site scripting

Cross-site scripting (XSS) involves injecting malicious code into an otherwise safe website. It is usually done through user input that is not sufficiently sanitised before being processed and stored on the server.

Students should be able to interpret fragments of JavaScript related to cross-site scripting.

Cascading Style Sheets (CSS)

Cascading style sheets (CSS) are used to describe the formatting of web pages. Students are expected to interpret code fragments written in CSS and HTML. The following syntax is used.

```
/* Comment */
```

```
selector {  
    property: value;  
}
```

- **Comments:** can be specified anywhere in CSS and are enclosed in /* and */
- **Selector:** targets a particular HTML element on the page to which the styling within the braces should be applied
- **Property:** a styling property, such as color
- **Value:** the value of the styling property, such as red

The following HTML fragment shows the CSS to style a webpage.

```
<html>  
  <head><title>My Website</title></head>  
  <body>  
    <h1>Welcome!</h1>  
    <p id="welcome">Welcome to my website!</p>  
    <p class="red-text">This text should be red</p>  
    <p>My website also has <span class="red-text">red text</span> here</p>  
  </body>  
</html>
```

The CSS to style that page is as follows.

```
/* This selector targets an HTML element */  
h1 {  
    font-size: 18px;  
}
```

```
/* This selector targets an HTML element with a specific "id" */  
#welcome {  
    font-style: italic;  
}
```

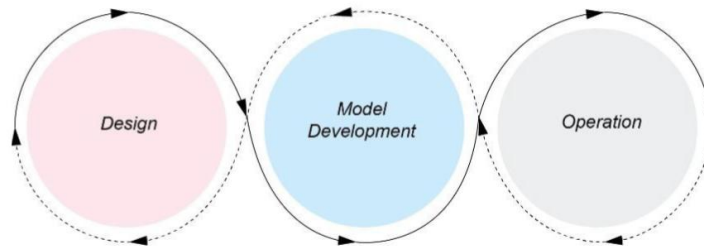
```
/* This selector targets an HTML element with a specific "class" */  
.red-text {  
    color: red;  
}
```

Machine Learning

Machine learning automation

MLOps is the automated process of designing, training and deploying machine learning models. It borrows many of the same principles and practices used in DevOps, bringing together the teams involved in developing machine learning models and the operational teams involved in deploying and supporting the models in production.

Students should know the three stages of MLOps.



- Design:
 - defining the business problem to be solved
 - refactoring the business problem into a machine learning problem
 - defining success metrics
 - researching available data.
- Model development:
 - data wrangling
 - feature engineering
 - model training
 - model testing and validation.
- Operations:
 - model deployment
 - supporting operations/use
 - monitoring model performance.

Regression algorithms

Linear regression and polynomial regression algorithms are used to predict values in a continuous range, such as integers. These regression algorithms are used for machine learning.

Logistic regression is used for classification problems.

Students should know how to design programs which use and apply these algorithms but are not expected to implement (or code) these complex algorithms.

The following Python code represents linear regression using NumPy and Scikit-learn machine learning frameworks.

```
# Import frameworks

import numpy as np

from sklearn.linear_model import LinearRegression

# Create the data for the two features

x = np.array([[2], [4], [6], [8], [10], [12], [14], [16]])
y = np.array([1, 3, 5, 7, 9, 11, 13, 15])

# Create the model

model = LinearRegression()

# Fit the model to the data (that is determine the line of best fit through the data)

model.fit(x, y)

# We can now use the model for predictions with existing or new data

# The model expects a value for x and will predict a value for y – so if we asked for a
prediction on 4 it would return 3 (as that is known data).

y_prediction = model.predict([[4]])

# This should print 3

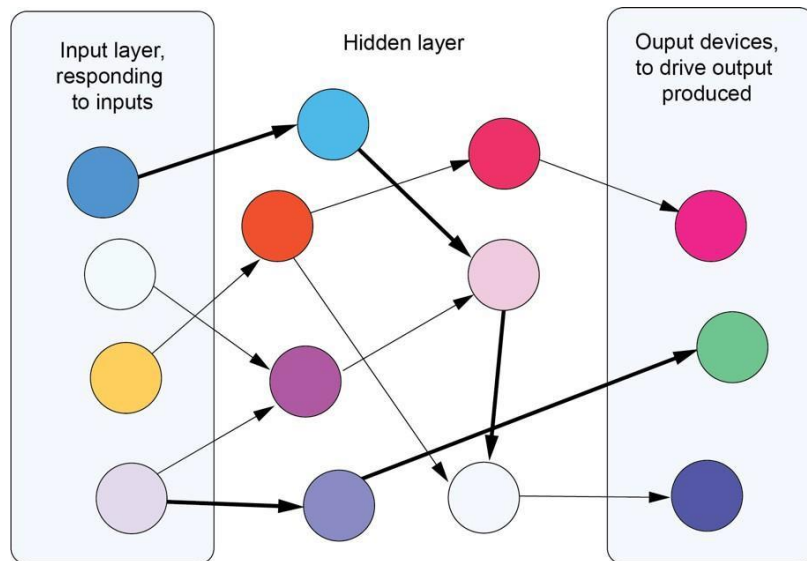
print(y_prediction)

# If we asked for a prediction of 4.5 it should return 3.5 (even though that is unknown data
we can tell what it should return, given there is a linear relationship)

y_prediction = model.predict([[4.5]])

# This should print 3.5
print(y_prediction)
```

Neural Networks



Neural networks were designed to mimic the processing inside the human brain. They consist of a series of interconnected nodes (artificial neurones). Each neurone can accept a binary input signal and potentially output another signal to connected nodes.

Training cycle

Internal weightings and threshold values for each node are determined in the initial training cycle for each neural network. The system is exposed to a series of inputs with known responses. Linear regression with backward chaining is used to iteratively determine the set of unique values required for output. Regular exposure to the training cycle results in improved accuracy and pattern matching.

Execution cycle

In the diagram, signal strength between nodes with the strongest weightings are thicker representing a higher priority in determining the final output. The execution cycle follows the training cycle and utilises the internal values developed during the training cycle to determine the output.

Methods for Testing a System

Students are expected to know and understand the following methods for testing a software solution.

- functional testing
- acceptance testing
- live data
- simulated data
- beta testing
- volume testing.

Character Representation

Characters can be represented with ASCII or Unicode. When a developer is working with text strings, they can make use of how the text data is stored internally in its binary format to perform functions such as changing the case of letters in the string, or performing a simple encryption.

Programming with Python

Students are expected to be able to code using the Python programming language.

Students should be familiar with the use of the following features:

- control structures
- global and local variables
- use of simple and structured data types
- classes, objects, attributes and methods
- functions
- modules and libraries
- file handling

Students are expected to design and implement programs incorporating combinations of these features.